

UNIT-3

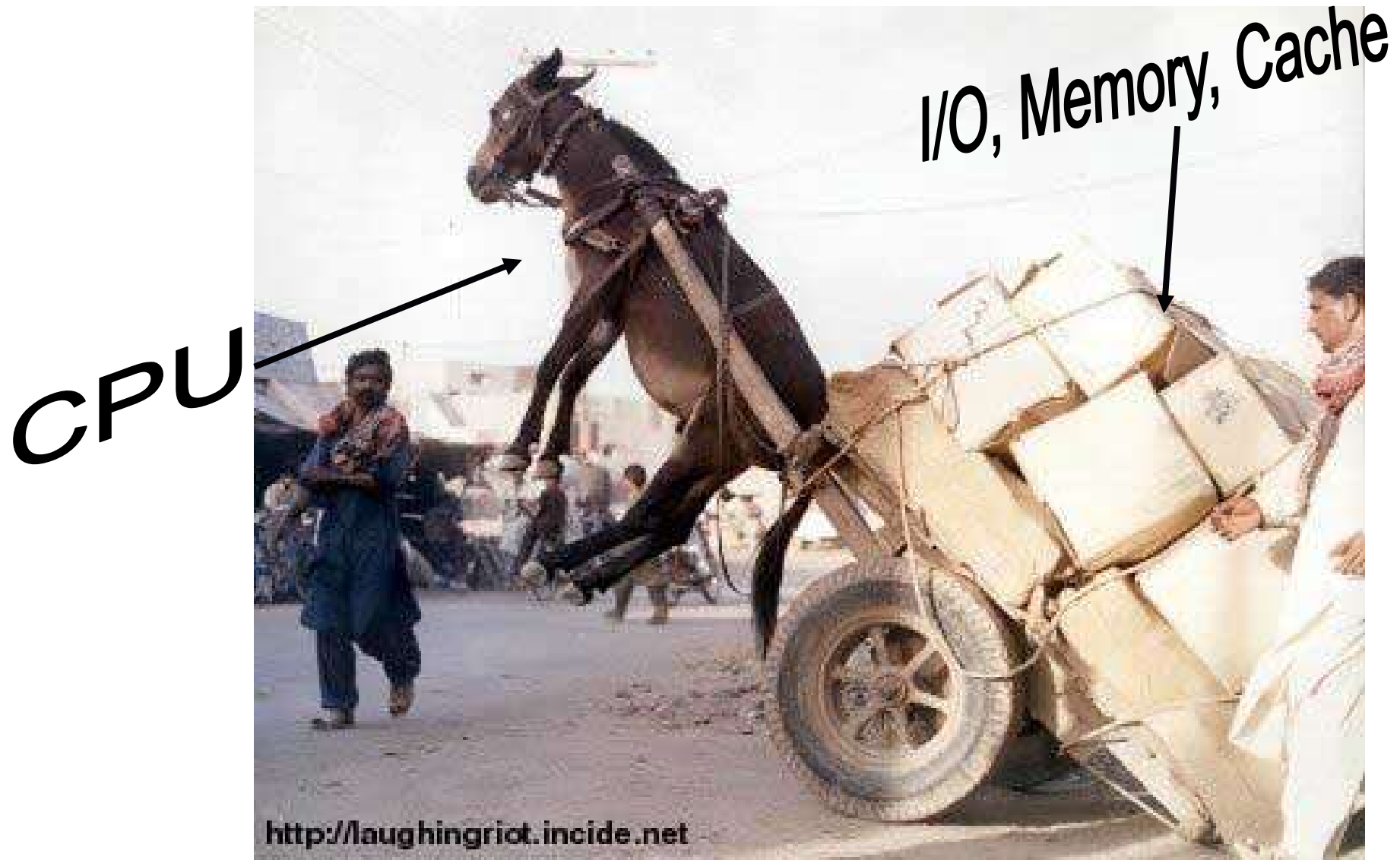
24CSEN2021:COMPUTER ORGANIZATION AND ARCHITECTURE

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Introduction

- Even a sophisticated processor may perform well below an ordinary processor:
 - Unless supported by matching performance by the memory system.
 - Unfortunately the gap is widening.
- The focus of this module:
 - Study how memory system performance has been enhanced through various innovations and optimizations.

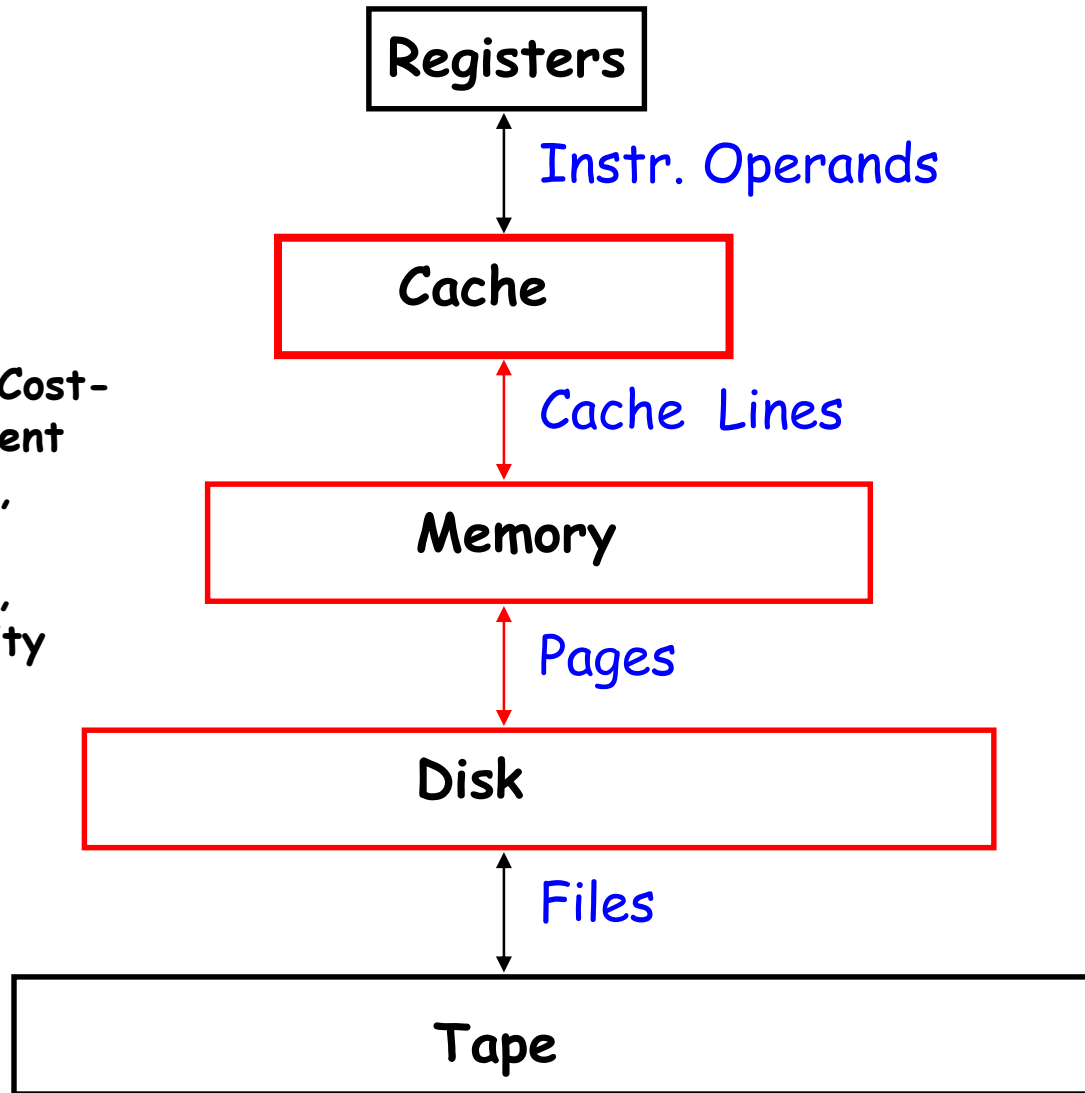
An Unbalanced System



Source: Bob Colwell keynote ISCA'29 2002

Levels of the Memory Hierarchy

Adv: Faster Access, Cost-Effective, Efficient use of resources, Increases the processing speed, Increased capacity

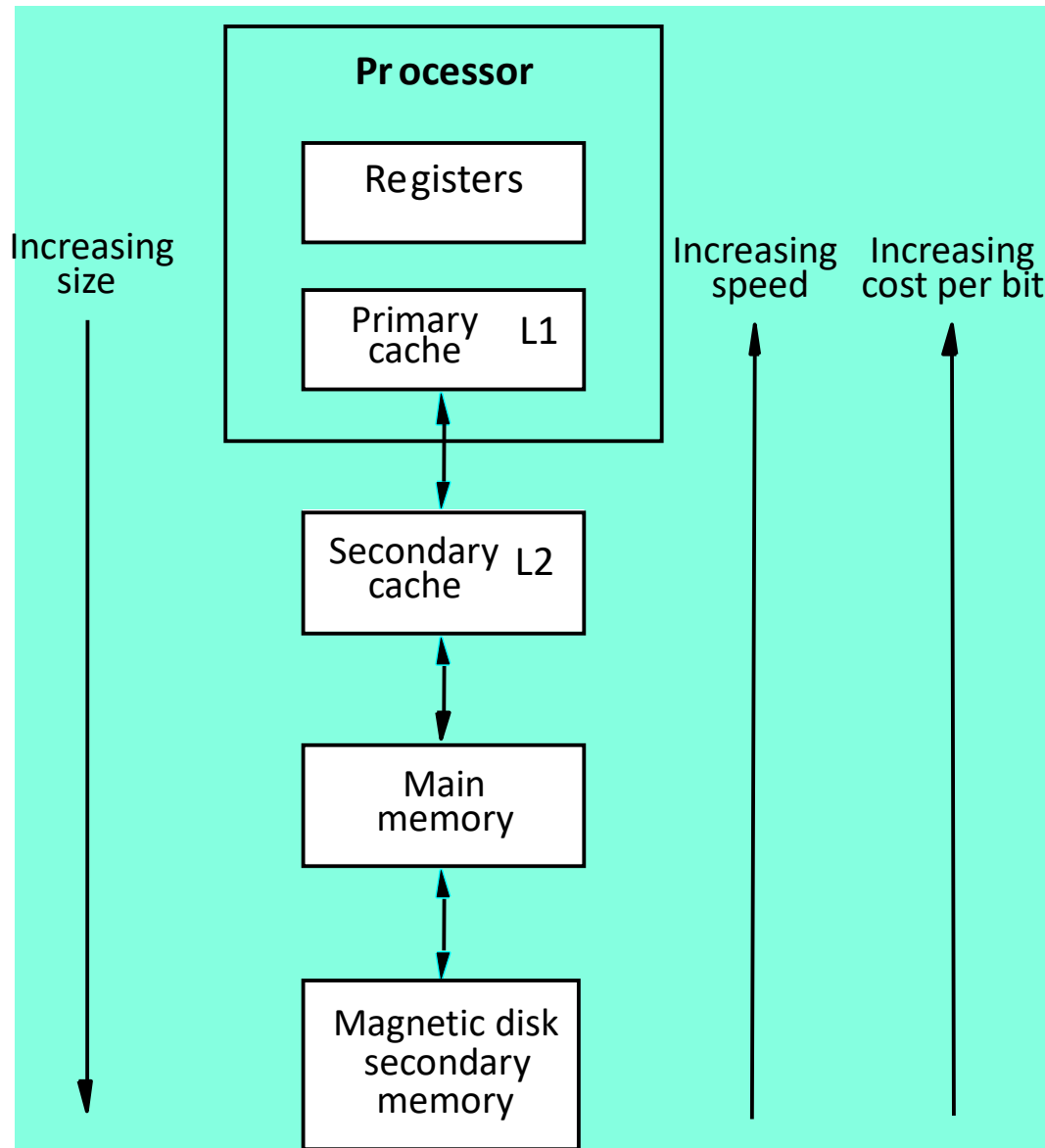


Upper Level
faster

Lower Level
4

Larger

Memory Hierarchy



- Fastest access is to the data held in processor registers. Registers are at the top of the memory hierarchy.
- Relatively small amount of memory that can be implemented on the processor chip. This is processor cache.
- Two levels of cache. Level 1 (L1) cache is on the processor chip. Level 2 (L2) cache is in between main memory and processor.
- Next level is main memory, implemented as SIMMs. Much larger, but much slower than cache memory.
- Next level is magnetic disks. Huge amount of inexpensive storage.
- Speed of memory access is critical, the idea is to bring instructions and data that will be used in the near future as close to the processor as possible.

Basic concepts

- Maximum size of the memory depends on the addressing scheme:
 - 16-bit computer generates 16-bit addresses and can address up to 2^{16} memory locations.
 - Number of locations represents the size of the address space of a computer.
- Most modern computers are byte-addressable.
- Memory is designed to store and retrieve data in word-length quantities.
 - Word length of a computer is commonly defined as the number of bits stored and retrieved in one memory operation.

Shared Memory Vs Distributed Memory:

- **Shared Memory:**

- Processors communicate through a shared memory.
- Typically processors connected to each other and to the shared memory through a bus.

- **Distributed Memory:**

- Processors do not share any physical memory.
- Processors connected to each other through a network.

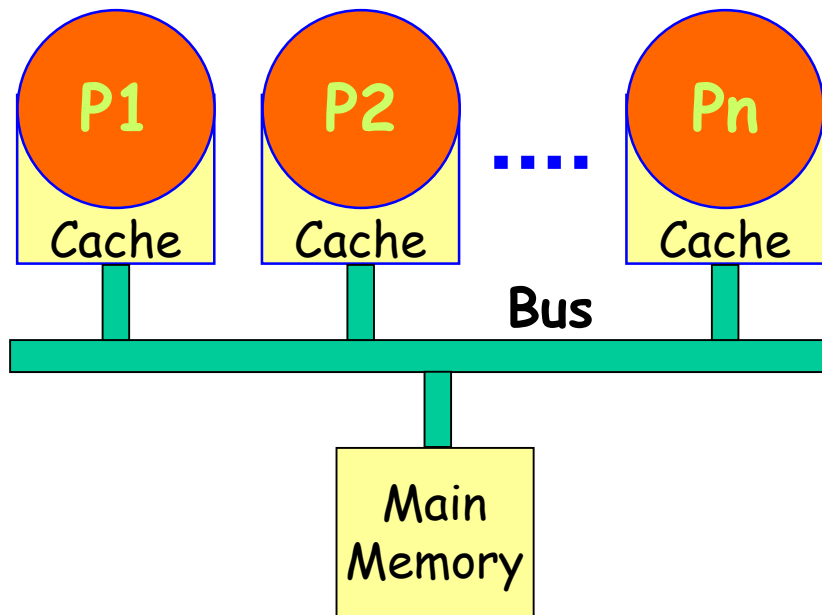
Shared Memory

- Shared memory located at a centralized location:
 - May consist of several interleaved modules --- same distance (access time) from any processor.
 - Also called Uniform Memory Access (UMA) model.

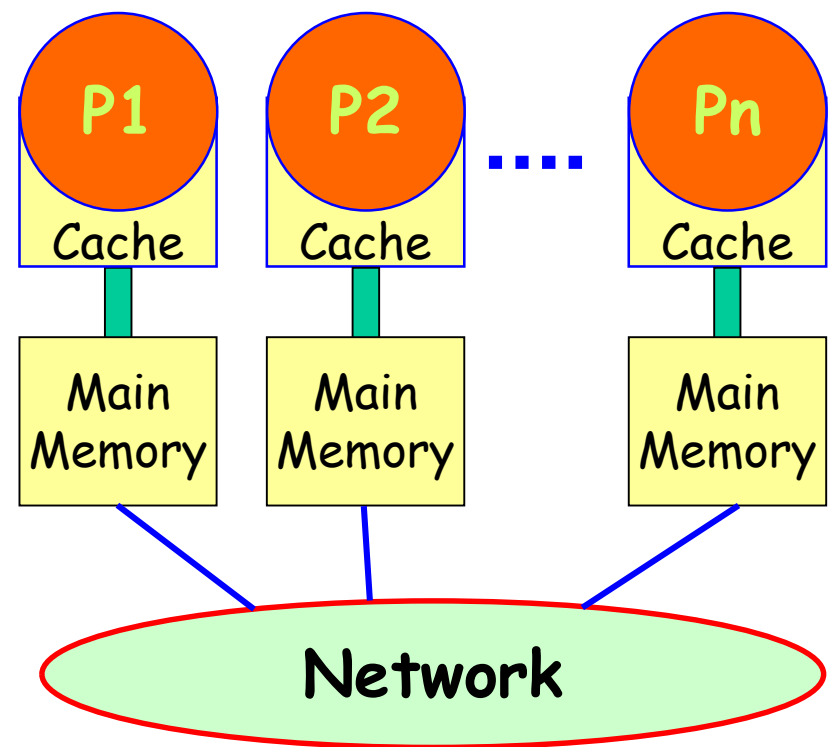
Distributed Memory

- Memory is distributed to each processor:
 - Improves scalability.
- Non-Uniform Memory Access (NUMA)
 - (a) Message passing architectures – No processor can directly access another processor's memory.
 - (b) Distributed Shared Memory (DSM)– Memory is distributed, but the address space is shared.

UMA vs. NUMA Computers

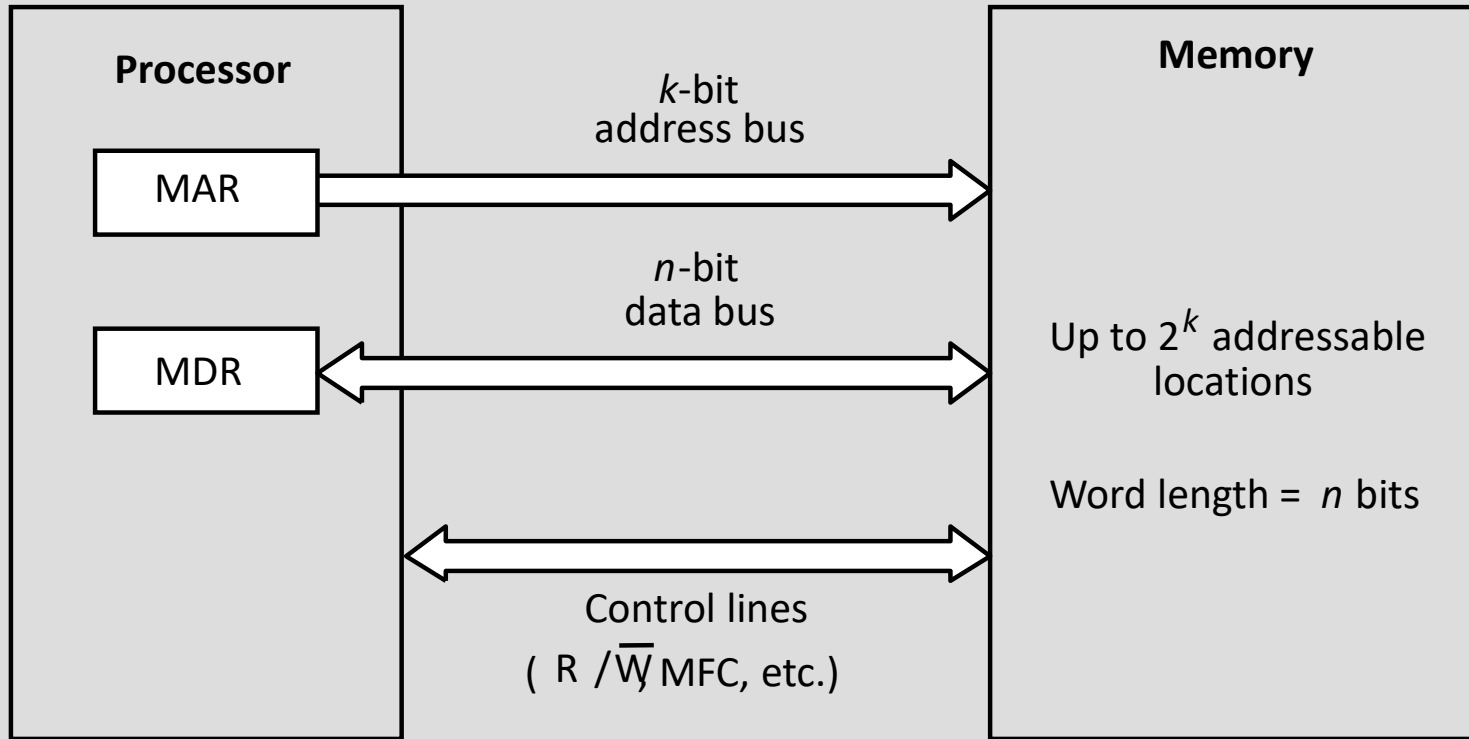


(a) UMA Model



(b) NUMA Model

Basic concepts (contd..)



Recall that the data transfers between a processor and memory involves two registers MAR and MDR.

If the address bus is k -bits, then the length of MAR is k bits.

If the word length is n -bits, then the length of MDR is n bits.

Control lines include R/W and MFC .

For Read operation $R/W = \underline{1}$ and for Write operation $R/W = \underline{0}$.

Basic concepts (contd..)

- Measures for the speed of a memory:
 - Elapsed time between the initiation of an operation and the completion of an operation is the memory access time.
 - Minimum time between the initiation of two successive memory operations is memory cycle time.
- In general, the faster a memory system, the costlier it is and the smaller it is.
- An important design issue is to provide a computer system with as large and fast a memory as possible, within a given cost target.
- Several techniques to increase the effective size and speed of the memory:
 - Cache memory (to increase the effective speed).
 - Virtual memory (to increase the effective size).

Semiconductor Memory

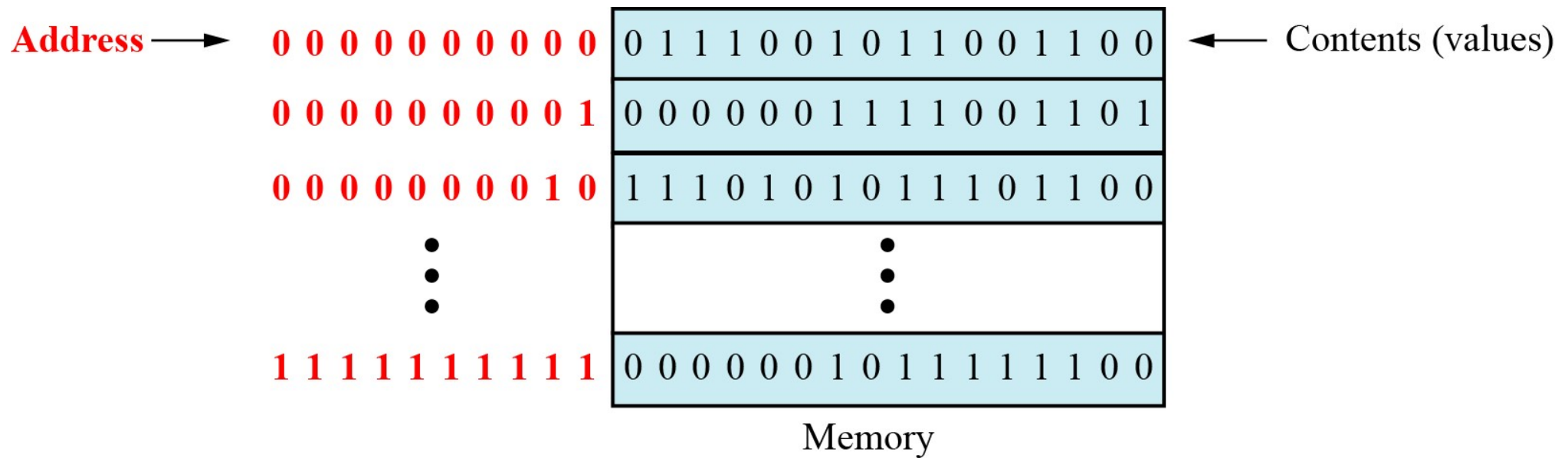
- RAM
 - Read/Write
 - Volatile
 - Temporary storage
 - Static or dynamic

Semiconductor RAM memories

- Random Access Memory (RAM) memory unit is a unit where any location can be addressed in a fixed amount of time, independent of the location's address.
- Internal organization of memory chips:
 - Each memory cell can hold one bit of information.
 - Memory cells are organized in the form of an array.
 - One row is one memory word.
 - All cells of a row are connected to a common line, known as the "word line".
 - Word line is connected to the address decoder.
 - Sense/write circuits are connected to the data input/output lines of the memory chip.

MEMORY LOCATIONS AND ADDRESSES

- **Main memory** is the second major **subsystem** in a computer. It consists of a collection of **storage locations**, each with a unique identifier, called an **address**.
- Data is transferred to and from memory in groups of bits called **words**. A word can be a group of 8 bits, 16 bits, 32 bits or 64 bits (and growing).
- If the word is 8 bits, it is referred to as a **byte**. The term “byte” is so common in computer science that sometimes a 16-bit word is referred to as a 2-byte word, or a 32-bit word is referred to as a 4-byte word.



Main memory

Address space

- To access a word in memory requires an identifier. Although programmers use a name to identify a word (or a collection of words), at the hardware level each word is identified by an address.
- The total **number** of **uniquely identifiable locations** in memory is called the **address space**.
- For example, a memory with 64 kilobytes (16 address lines required) and a word size of 1 byte has an address space that ranges from **0 to 65,535**.

Table 5.1 Memory units

<i>Unit</i>	<i>Exact Number of Bytes</i>	<i>Approximation</i>
kilobyte	2^{10} (1024) bytes	10^3 bytes
megabyte	2^{20} (1,048,576) bytes	10^6 bytes
gigabyte	2^{30} (1,073,741,824) bytes	10^9 bytes
terabyte	2^{40} bytes	10^{12} bytes



Memory addresses are defined using unsigned binary integers.

Example 1

A computer has 32 MB (megabytes) of memory. How many bits are needed to address any single byte in memory?

Example 2

A computer has 128 MB of memory. Each word in this computer is eight bytes. How many bits are needed to address any single word in memory?

Example 1

A computer has 32 MB (megabytes) of memory. How many bits are needed to address any single byte in memory?

Solution

The memory address space is 32 MB, or 2^{25} ($2^5 \times 2^{20}$). This means that we need $\log_2 2^{25}$, or **25 bits**, to address each byte.

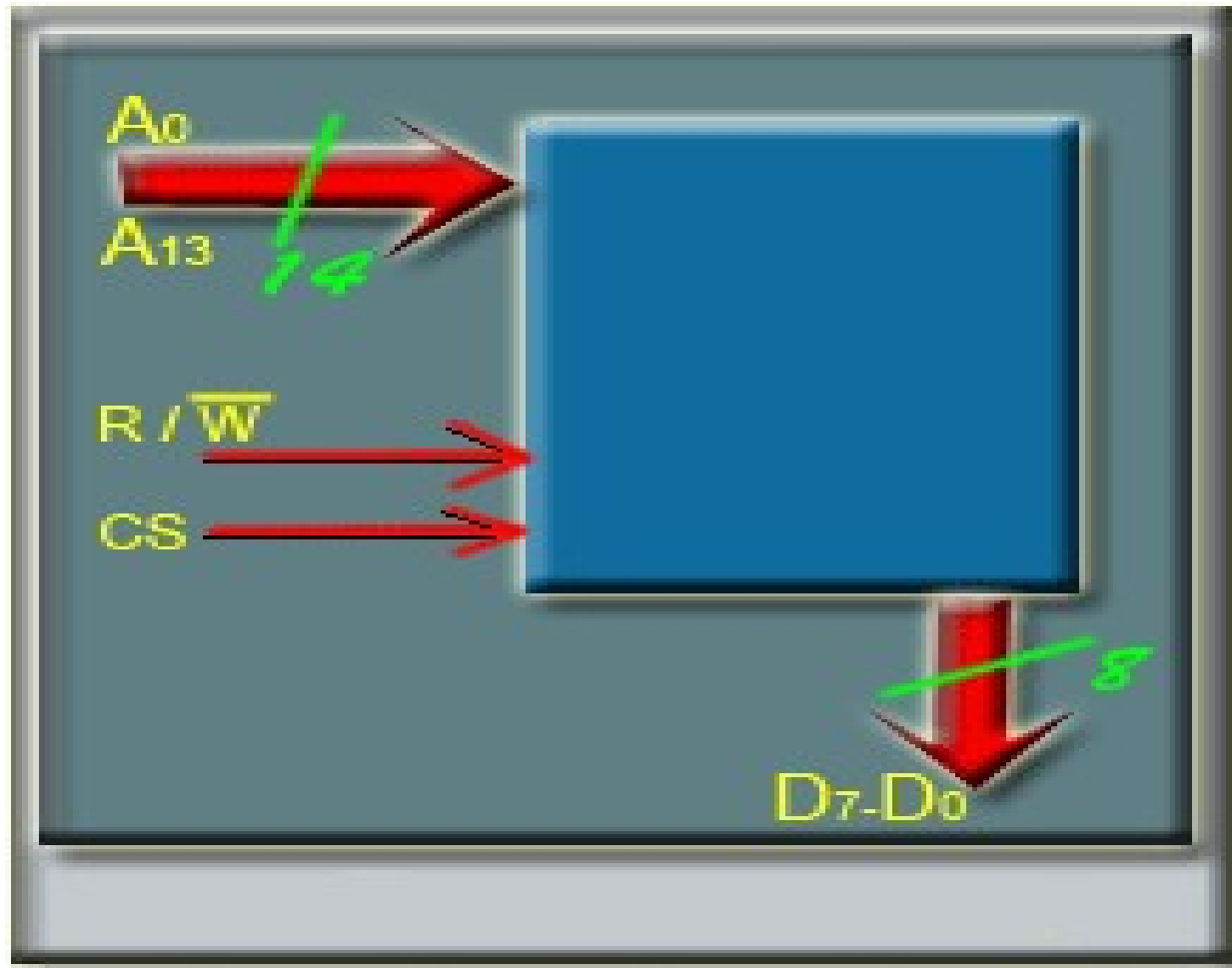
Example 2

A computer has 128 MB of memory. Each word in this computer is eight bytes. How many bits are needed to address any single word in memory?

Solution

The memory address space is 128 MB, which means 2^{27} . However, each word is eight (2^3) bytes, which means that we have 2^{24} words. This means that we need $\log_2 2^{24}$, or **24 bits**, to address each word.

16k x 8 Memory chip



128 x 16 RAM Memory chip
512 X 32 RAM Memory chip
1k X 16 RAM chip

SRAM v DRAM

- Both volatile
 - Power needed to preserve data
- Dynamic cell
 - Simpler to build, smaller
 - More dense
 - Less expensive
 - Needs refresh
 - Larger memory units
- Static
 - Faster

Semiconductor RAM memories (contd..)

- Static RAMs (SRAMs):

- Consist of **circuits** that are capable of retaining their state as long as the power is applied.
- Volatile memories, because their contents are lost when power is interrupted.
- Access times of static RAMs are in the range of few nanoseconds.
- However, the **cost** is usually **high**.

- Dynamic RAMs (DRAMs):

- Do not retain their state indefinitely.
- Contents must be periodically refreshed.
- Contents may be refreshed while accessing them for reading.

Read Only Memory (ROM)

- Permanent storage
 - Nonvolatile
- Library subroutines
- Systems programs (BIOS)
- Function tables

Key Differences Between RAM and ROM

Feature	RAM	ROM
Volatility	Volatile (loses data when powered off)	Non-volatile (retains data when powered off)
Read/Write	Can be read from and written to	Primarily read-only
Speed	Faster	Slower than RAM, but faster than hard drives
Capacity	Generally larger (measured in GB)	Typically smaller (measured in MB)
Usage	Temporary data storage for active processes	Permanent storage for essential instructions
Cost	More expensive per GB	Less expensive per GB
Data Retention	Short-term	Long-term

Flash memory

- Flash memory has similar approach to EEPROM.
- Read the contents of a single cell, but write the contents of an entire block of cells.
- Flash devices have greater density.
 - Higher capacity and low storage cost per bit.
- Power consumption of flash memory is very low, making it attractive for use in equipment that is battery-driven.
- Single flash chips are not sufficiently large, so larger memory modules are implemented using flash cards and flash drives.

Semiconductor Memory Types

Memory Type	Category	Erasure	Write Mechanism	Volatility
Random-access memory (RAM)	Read-write memory	Electrically, byte-level	Electrically	Volatile
Read-only memory (ROM)	Read-only memory	Not possible	Masks	Nonvolatile
Programmable ROM (PROM)				
Erasable PROM (EPROM)	Read-mostly memory	UV light, chip-level	Electrically	
Electrically Erasable PROM (EEPROM)		Electrically, byte-level		
Flash memory		Electrically, block-level		

Acronyms

RAM	--Random Access Memory	time taken to access any arbitrary location in memory is constant (c.f., disks)
ROM	--Read Only Memory	ROMs are RAMs which can only be written to once; thereafter they can only be read Older uses included storage of bootstrap info
PROM	--Programmable ROM	A ROM which can be bench programmed
EPROM	--Erasable PROM	A PROM which can be erased for rewriting
EEPROM	--Electrically EPROM	A PROM which can be erased electrically.
SRAM	--Static RAM	RAM chip which loses contents upon power off
DRAM	--Dynamic RAM	RAM chip whose contents need to be refreshed.
SDRAM	--Synchronous DRAM	RAM whose memory operations are synchronized with a clock signal.

Types of ROM

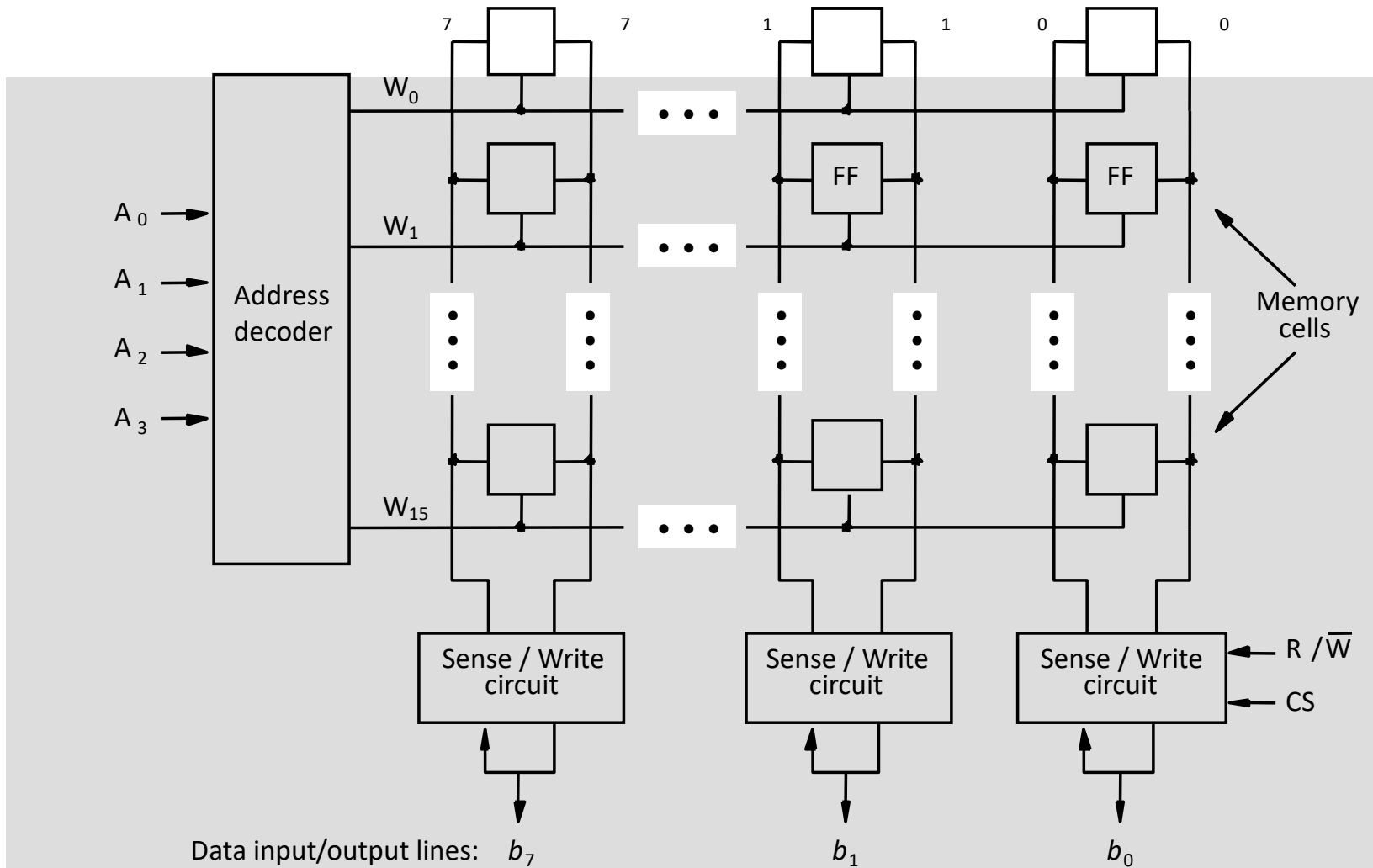
- Written during manufacture
 - Very expensive for small runs
- Programmable (once)
 - PROM
 - Needs special equipment to program
- Read “mostly”
 - Erasable Programmable (EPROM)
 - Erased by UV
 - Electrically Erasable (EEPROM)
 - Takes much longer to write than read
 - Flash memory
 - Erase whole memory electrically

Organisation in detail

- A 16Mbit chip can be organised as 1M of 16 bit words
- A bit per chip system has 16 lots of 1Mbit chip with bit 1 of each word in chip 1 and so on
- A 16Mbit chip can be organised as a 2048 x 2048 x 4bit array
 - Reduces number of address pins
 - Multiplex row address and column address
 - 11 pins to address ($2^{11}=2048$)
 - Adding one more pin doubles range of values so x4 capacity

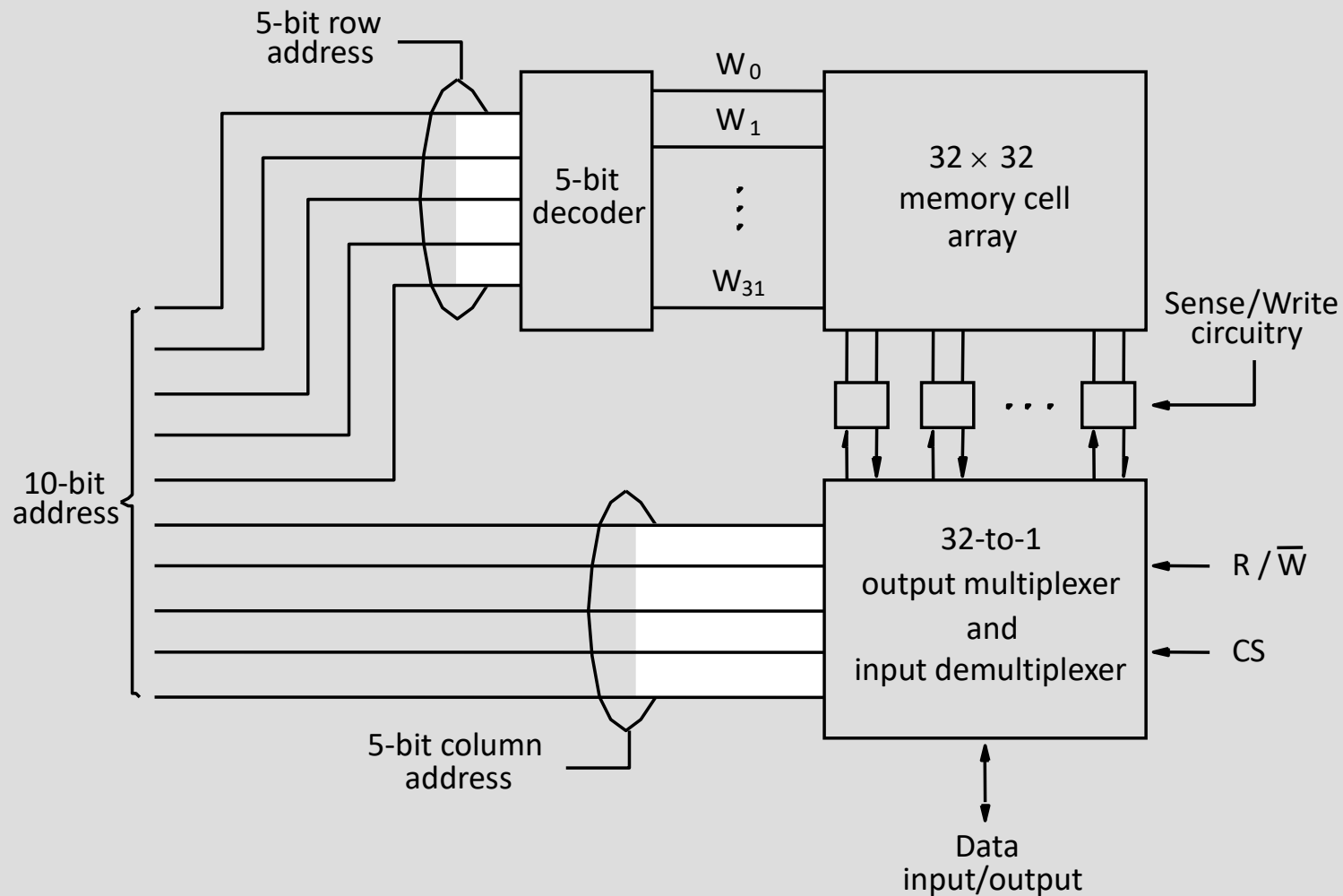
Semiconductor RAM memories (contd..)

Internal organization of memory chips (16 X 8 Memory Organization)



Semiconductor RAM memories (contd..)

Internal organization of memory chips



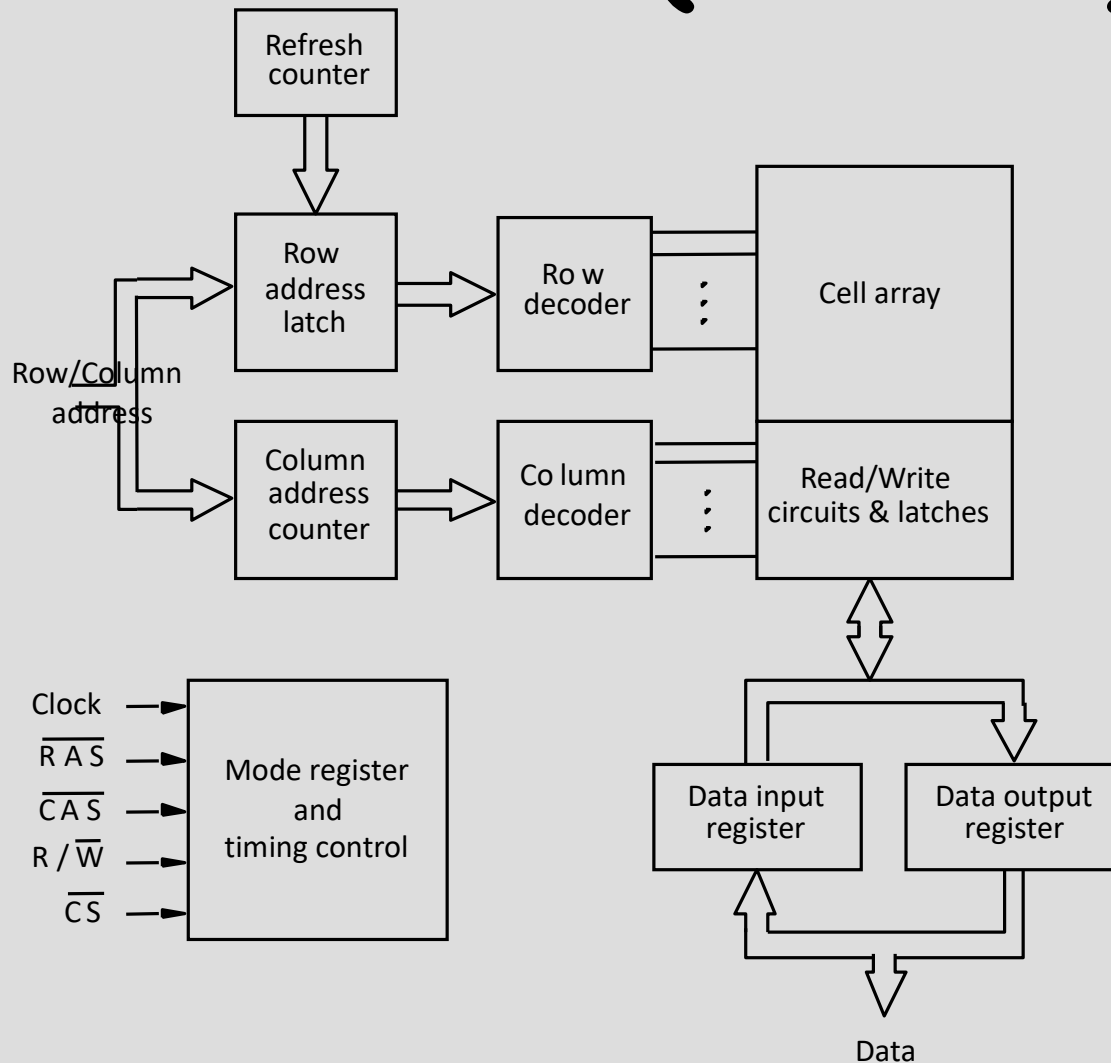
Semiconductor RAM memories (contd..)

- Recall the operation of the memory:
 - First all the contents of a row are selected based on a row address.
 - Particular byte is selected based on the column address.
- Suppose if we want to access the consecutive bytes in the selected row.
- This can be done without having to reselect the row.
 - Add a latch at the output of the sense circuits in each row.
 - All the latches are loaded when the row is selected.
 - Different column addresses can be applied to select and place different bytes on the data lines.

Semiconductor RAM memories (contd..)

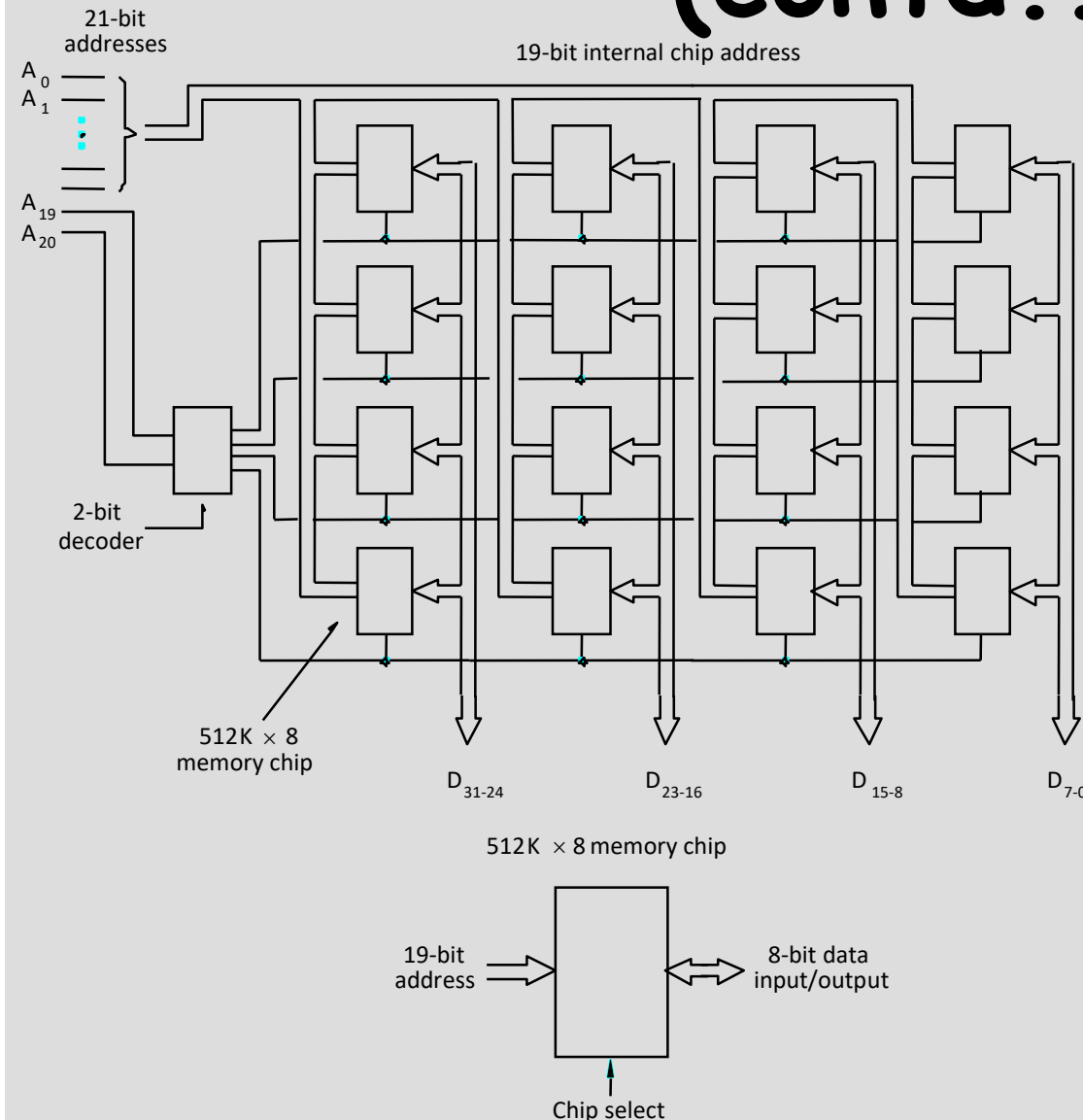
- Consecutive sequence of column addresses can be applied under the control signal **CAS**, without reselecting the row.
 - Allows a block of data to be transferred at a much faster rate than random accesses.
 - A small collection/group of bytes is usually referred to as a block.
- This transfer capability is referred to as the fast page mode feature.

Semiconductor RAM memories (contd..)



- Operation is directly synchronized with processor clock signal.
- Synchronous DRAMs.
- The outputs of the sense circuits are connected to a latch.
- During a Read operation, the contents of the cells in a row are loaded onto the latches.
- During a refresh operation, the contents of the cells are refreshed without changing the contents of the latches.
- Data held in the latches correspond to the selected columns are transferred to the output.
- For a burst mode of operation, successive columns are selected using column address counter and clock. CAS signal need not be generated externally.

Semiconductor RAM memories (contd..)



Implement a memory unit of $2M$ words of 32 bits each.
 Use $512K \times 8$ static memory chips.
 Each column consists of 4 chips.
 Each chip implements one byte position.
 A chip is selected by setting its chip select control line to 1.
 Selected chip places its data on the data output line, outputs of other chips are in high impedance state.
 21 bits to address a 32-bit word.
 High order 2 bits are needed to select the row, by activating the four Chip Select signals.
 19 bits are used to access specific byte locations inside the selected chip.

Semiconductor RAM memories (contd..)

- Large dynamic memory systems can be implemented using DRAM chips in a similar way to static memory systems.
- Placing large memory systems directly on the motherboard will occupy a large amount of space.
 - Also, this arrangement is inflexible since the memory system cannot be expanded easily.
- Packaging considerations have led to the development of larger memory units known as SIMMs (Single In-line Memory Modules) and DIMMs (Dual In-line Memory Modules).
- Memory modules are an assembly of memory chips on a small board that plugs vertically onto a single socket on the motherboard.
 - Occupy less space on the motherboard.
 - Allows for easy expansion by replacement.

Semiconductor RAM memories (contd..)

- Recall that the rows of the Dynamic RAM need to be accessed periodically in order to be refreshed.
- Older DRAMs typical refreshing period was 16 ms.
- SDRAMs typical refreshing period is 64 ms.
- Let us consider a SDRAM with 8192 rows:
 - Suppose it takes 4 clock cycles to access each row.
 - Total of $4 \times 8192 = 32768$ clock cycles.
 - If the clock rate is 133 MHz, then the total refresh time is 0.000246 seconds.
 - Refreshing must be done every 64 ms.
 - Overhead is 0.0038.

Semiconductor RAM memories

(contd..)

- Data is transferred between the processor and the memory in units of single word or a block.
- Speed and efficiency of data transfer has a large impact on the performance of a computer.
- Two metrics of performance: Latency and Bandwidth.
- Latency:
 - Time taken to transfer a single word of data to or from memory.
 - Definition is clear if the memory operation involves transfer of a single word of data.
 - In case of a block transfer, latency is the time it takes to transfer first word of data.
 - Time required to transfer first word in a block is substantially larger than the time required to transfer consecutive words in a block.

Semiconductor RAM memories (contd..)

- How much time is needed to transfer a single block of data.
- Blocks can be variable in size, it is useful to define a performance measure in terms of the number of bits or bytes transferred in one second.
- This performance measure is referred to as memory bandwidth.
- Bandwidth of a memory unit depends on:
 - Speed of access to the chip.
 - How many bits can be accessed in parallel.
- Bandwidth of data transfer between processor and memory unit also depends on:
 - Transfer capability of the links that connect the processor and memory, or the speed of the bus.

Memory systems

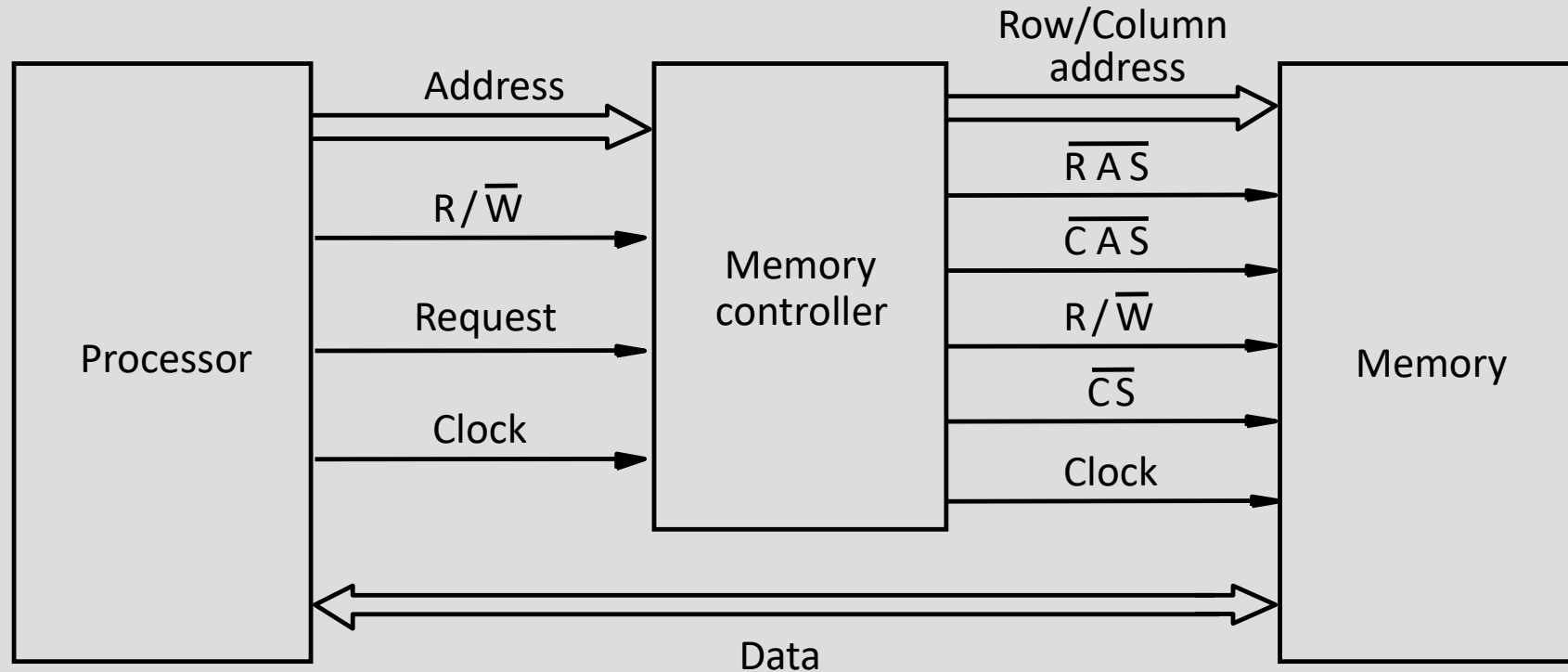
- Various factors such as cost, speed, power consumption and size of the chip determine how a RAM is chosen for a given application.
- Static RAMs:
 - Chosen when speed is the primary concern.
 - Circuit implementing the basic cell is highly complex, so cost and size are affected.
 - Used mostly in cache memories.
- Dynamic RAMs:
 - Predominantly used for implementing computer main memories.
 - High densities available in these chips.
 - Economically viable for implementing large memories.

Memory controller

- Recall that in a dynamic memory chip, to reduce the number of pins, multiplexed addresses are used.
- Address is divided into two parts:
 - High-order address bits select a row in the array. _____
 - They are provided first, and latched using RAS signal.
 - Low-order address bits select a column in the row. _____
 - They are provided later, and latched using CAS signal.
- However, a processor issues all address bits at the same time.
- In order to achieve the multiplexing, memory controller circuit is inserted between the processor and memory.

• (Row Address Strobe)

Memory controller (contd..)



Memory controller accepts the complete address, $\overline{R/W}$ signal from the processor, under the request of a control signal.

Controller forwards row and column address portions to the memory, and issues $\overline{RAS}$ and $\overline{CAS}$ signals.

It also sends $R/\bar{W}$ and $\overline{CS}$ signals to the memory.

Data lines are connected directly between the processor and memory.

Read-Only Memories (ROMs)

- SRAM and SDRAM chips are volatile:
 - Lose the contents when the power is turned off.
- Many applications need memory devices to retain contents after the power is turned off.
 - For example, computer is turned on, the operating system must be loaded from the disk into the memory.
 - Store instructions which would load the OS from the disk.
 - Need to store these instructions so that they will not be lost after the power is turned off.
 - We need to store the instructions into a non-volatile memory.
- Non-volatile memory is read in the same manner as volatile memory.
 - Separate writing process is needed to place information in this memory.
 - Normal operation involves only reading of data, this type of memory is called Read-Only memory (ROM).

Read-Only Memories (contd..)

- Read-Only Memory:
 - Data are written into a ROM when it is manufactured.
- Programmable Read-Only Memory (PROM):
 - Allow the data to be loaded by a user.
 - Process of inserting the data is irreversible.
 - Storing information specific to a user in a ROM is expensive.
 - Providing programming capability to a user may be better.
- Erasable Programmable Read-Only Memory (EPROM):
 - Stored data to be erased and new data to be loaded.
 - Flexibility, useful during the development phase of digital systems.
 - Erasable, reprogrammable ROM.
 - Erasure requires exposing the ROM to UV light.

Read-Only Memories (contd..)

- Electrically Erasable Programmable Read-Only Memory (EEPROM):
 - To erase the contents of EPROMs, they have to be exposed to ultraviolet light.
 - Physically removed from the circuit.
 - EEPROMs the contents can be stored and erased electrically.

Flash memory

- Flash memory has similar approach to EEPROM.
- Read the contents of a single cell, but write the contents of an entire block of cells.
- Flash devices have greater density.
 - Higher capacity and low storage cost per bit.
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Speed, Size, and Cost

- A big challenge in the design of a computer system is to provide a sufficiently large memory, with a reasonable speed at an affordable cost.
- **Static RAM:**
 - Very fast, but expensive, because a basic SRAM cell has a complex circuit making it impossible to pack a large number of cells onto a single chip.
- **Dynamic RAM:**
 - Simpler basic cell circuit, hence are much less expensive, but significantly slower than SRAMs.
- **Magnetic disks:**
 - Storage provided by DRAMs is higher than SRAMs, but is still less than what is necessary.
 - Secondary storage such as magnetic disks provide a large amount of storage, but is much slower than DRAMs.

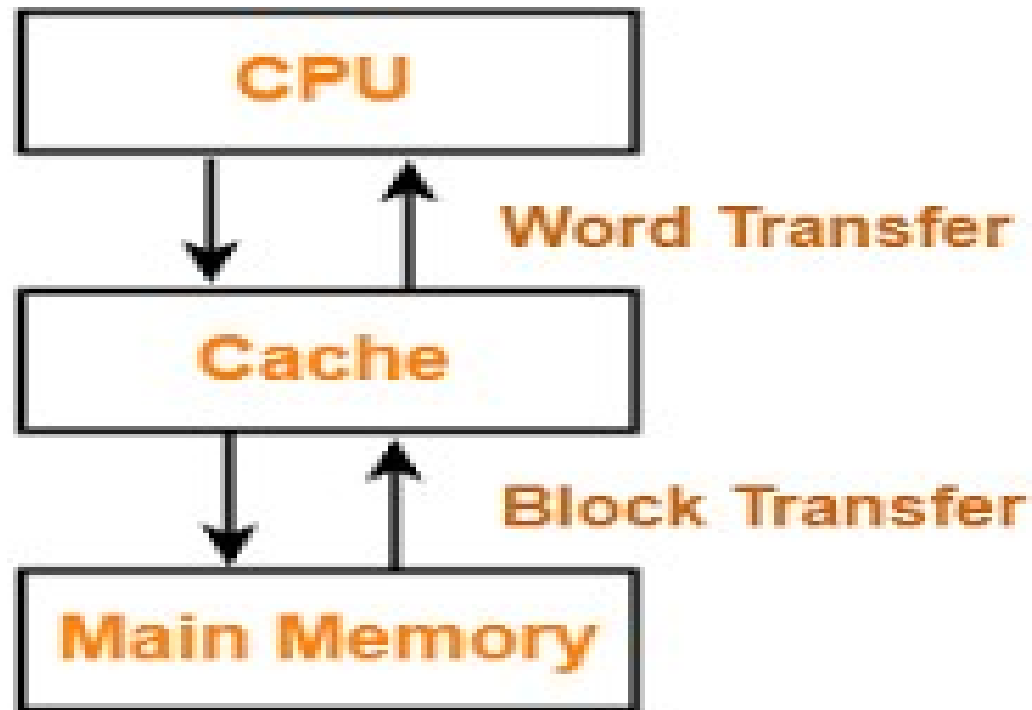
Cache Memories

The Memory System

Cache Memories

- Processor is much faster than the main memory.
 - As a result, the processor has to spend much of its time waiting while instructions and data are being fetched from the main memory.
 - Major obstacle towards achieving good performance.
- Speed of the main memory cannot be increased beyond a certain point.
- Cache memory is an architectural arrangement which makes the main memory appear faster to the processor than it really is.
- Cache memory is based on the property of computer programs known as "locality of reference".

Cache Memories



Cache and Main Memory

Locality of Reference

- Analysis of programs indicates that many instructions in localized areas of a program are executed repeatedly during some period of time, while the others are accessed relatively less frequently.
 - These instructions may be the ones in a loop, nested loop or few procedures calling each other repeatedly.
 - This is called "locality of reference".
- **Temporal locality of reference:**
 - Recently executed instruction is likely to be executed again very soon.
- **Spatial locality of reference:**
 - Instructions with addresses close to a recently instruction are likely to be executed soon.

1. Temporal Locality (Reusing Data)

```
#include <stdio.h>

int main() {
    int sum = 0;
    int arr[10] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};

    // This loop exhibits temporal locality for 'sum' and 'arr' elements
    for (int i = 0; i < 10; i++) {
        sum += arr[i]; // 'sum' is repeatedly accessed and modified
                     // 'arr[i]' is accessed in each iteration
    }

    printf("Sum: %d\n", sum);
    return 0;
}
```


2. Spatial Locality (Accessing Nearby Data)

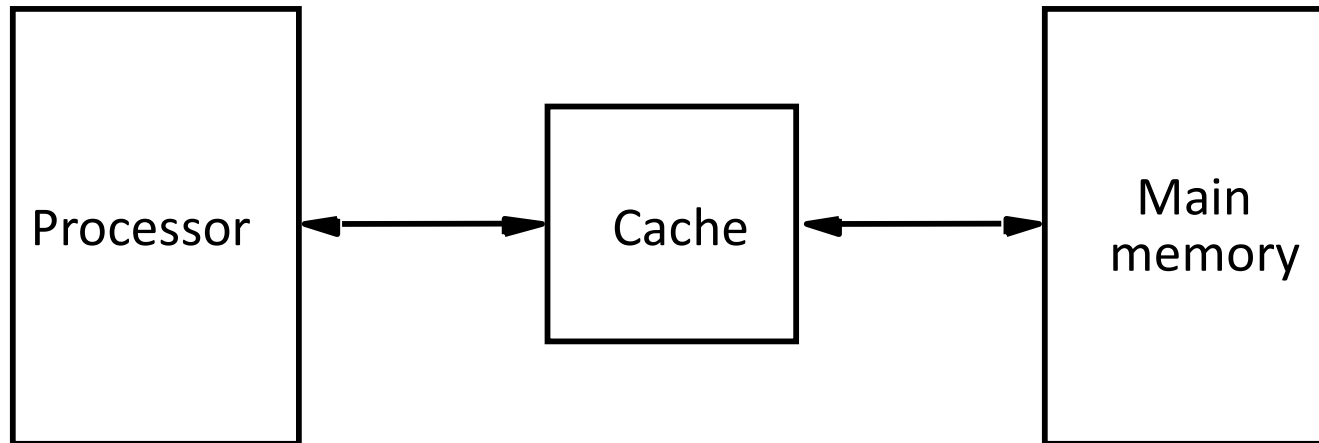
```
#include <stdio.h>

int main() {
    int matrix[3][3] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};

    // This nested loop exhibits spatial locality for 'matrix' elements
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            printf("%d ", matrix[i][j]); // Accessing elements in row-major order
        }
        printf("\n");
    }

    return 0;
}
```

Cache memories



- Processor issues a Read request, a block of words is transferred from the main memory to the cache, one word at a time.
- Subsequent references to the data in this block of words are found in the cache.
- At any given time, only some blocks in the main memory are held in the cache. Which blocks in the main memory are in the cache is determined by a "mapping function".
- When the cache is full, and a block of words needs to be transferred from the main memory, some block of words in the cache must be replaced. This is determined by a "replacement algorithm".

Cache hit

- Existence of a cache is transparent to the processor. The processor issues Read and Write requests in the same manner.
- If the data is in the cache it is called a Read or Write hit.
- Read hit:
 - The data is obtained from the cache.
- Write hit:
 - Cache has a replica of the contents of the main memory.
 - Contents of the cache and the main memory may be updated simultaneously. This is the write-through protocol.
 - Update the contents of the cache, and mark it as updated by setting a bit known as the dirty bit or modified bit. The contents of the main memory are updated when this block is replaced. This is write-back or copy-back protocol.

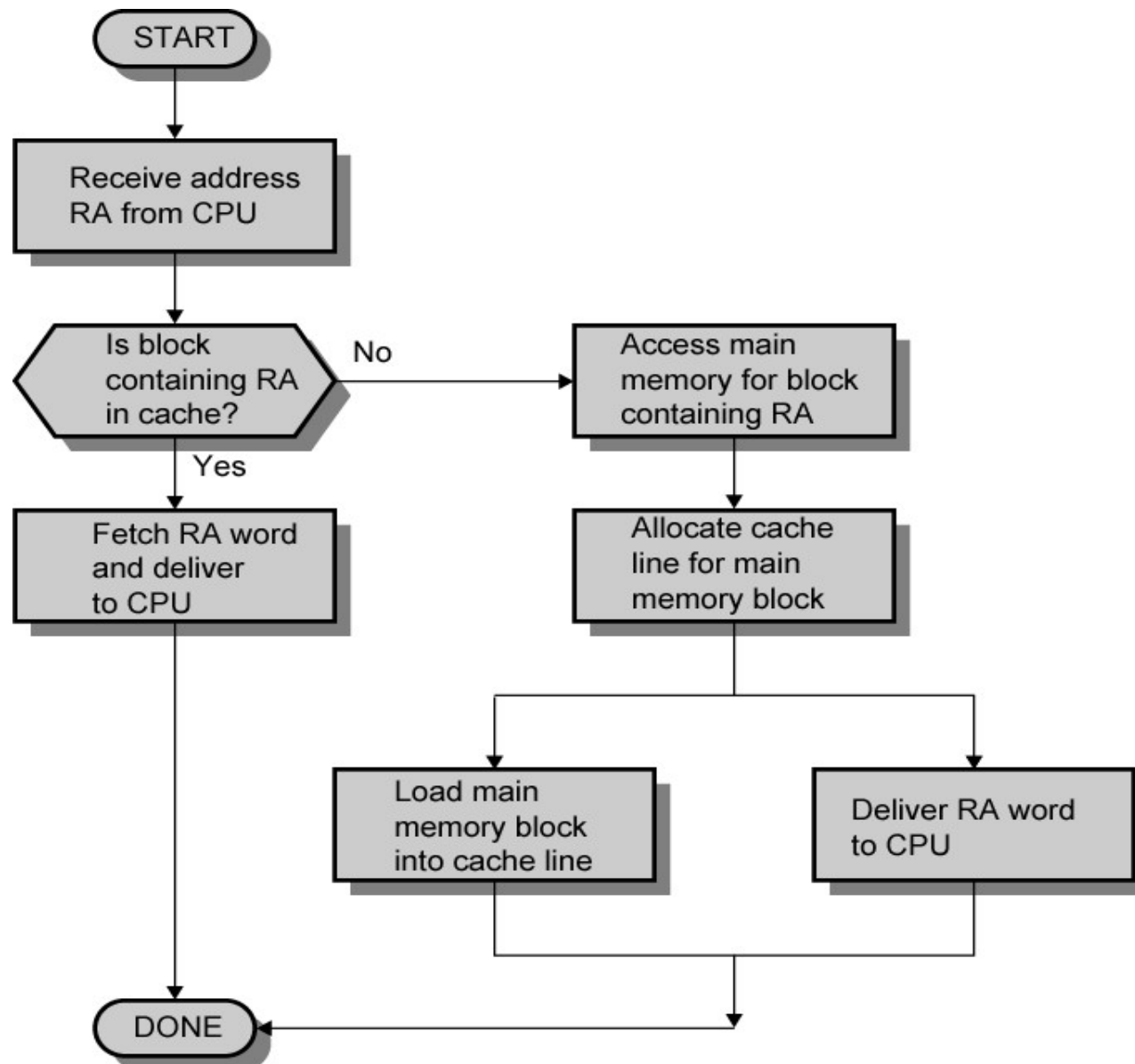
Cache miss

- If the data is not present in the cache, then a Read miss or Write miss occurs.
- Read miss:
 - Block of words containing this requested word is transferred from the memory.
 - After the block is transferred, the desired word is forwarded to the processor.
 - The desired word may also be forwarded to the processor as soon as it is transferred without waiting for the entire block to be transferred. This is called load-through or early-restart.
- Write-miss:
 - Write-through protocol is used, then the contents of the main memory are updated directly.
 - If write-back protocol is used, the block containing the addressed word is first brought into the cache. The desired word is overwritten with new information.

Cache operation – overview

- CPU requests contents of memory location
- Check cache for this data
- If present, get from cache (fast)
- If not present, read required block from main memory to cache
- Then deliver from cache to CPU
- Cache includes tags to identify which block of main memory is in each cache slot

Cache Read Operation - Flowchart



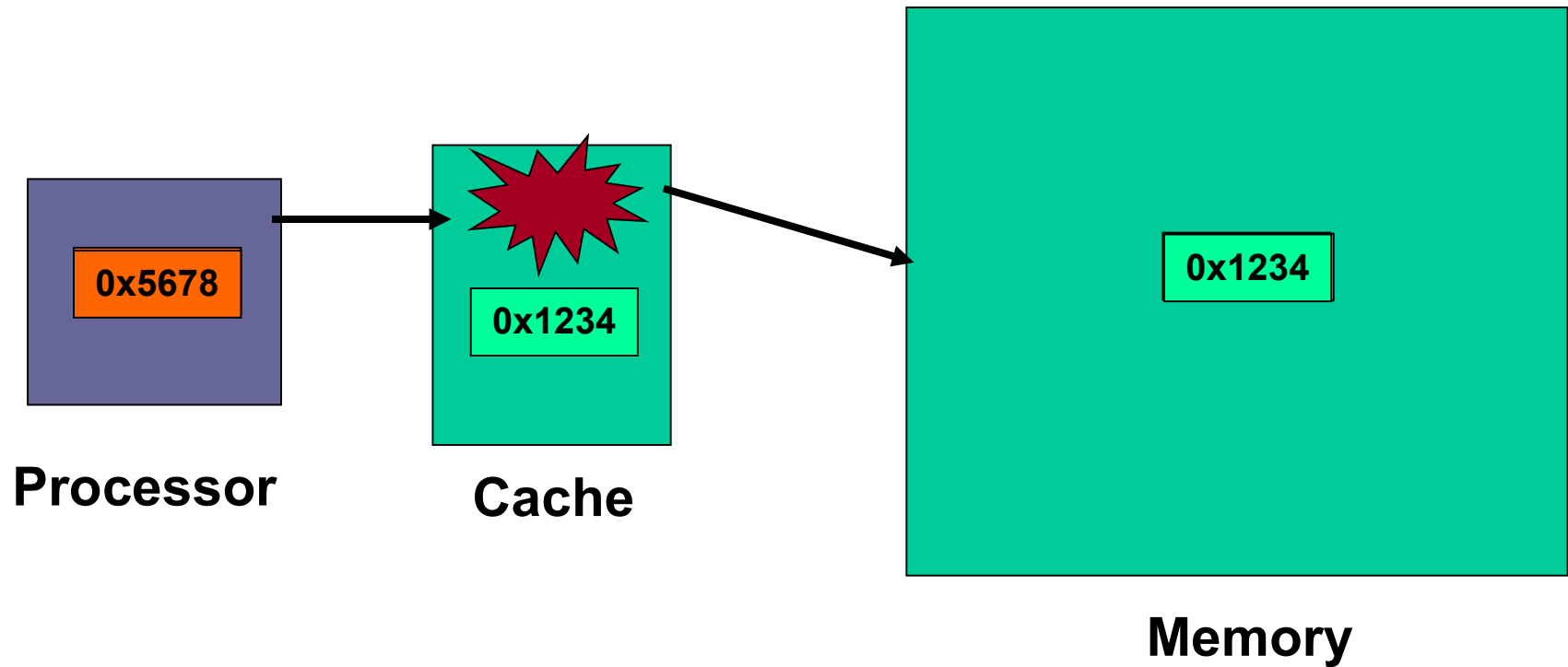
Cache Write Policies

- **Write-through:** Information is written to both the block in the cache and that in memory
- **Write-back:** Information is written back to memory only when a block frame is replaced:
 - Uses a “dirty” bit to indicate whether a block was actually written to,
 - Saves unnecessary writes to memory when a block is “clean”

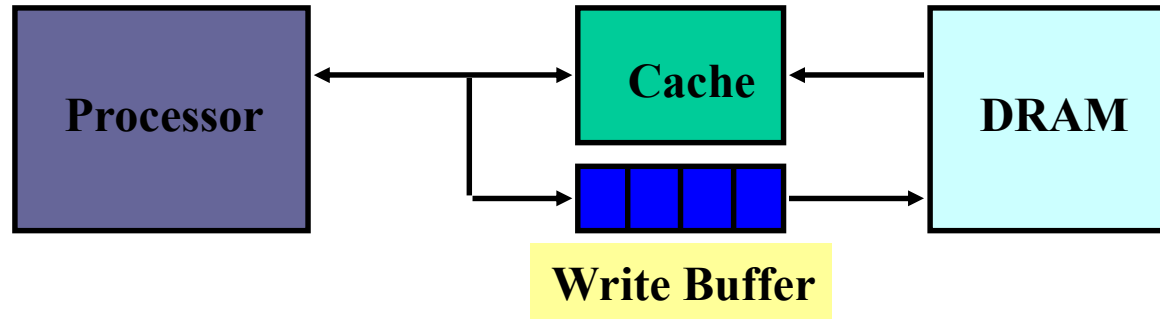
Trade-offs

- Write back
 - Faster because writes occur at the speed of the cache, not the memory.
 - Faster because multiple writes to the same block is written back to memory only once, uses less memory bandwidth.
- Write through
 - Easier to implement

Write-Through Policy

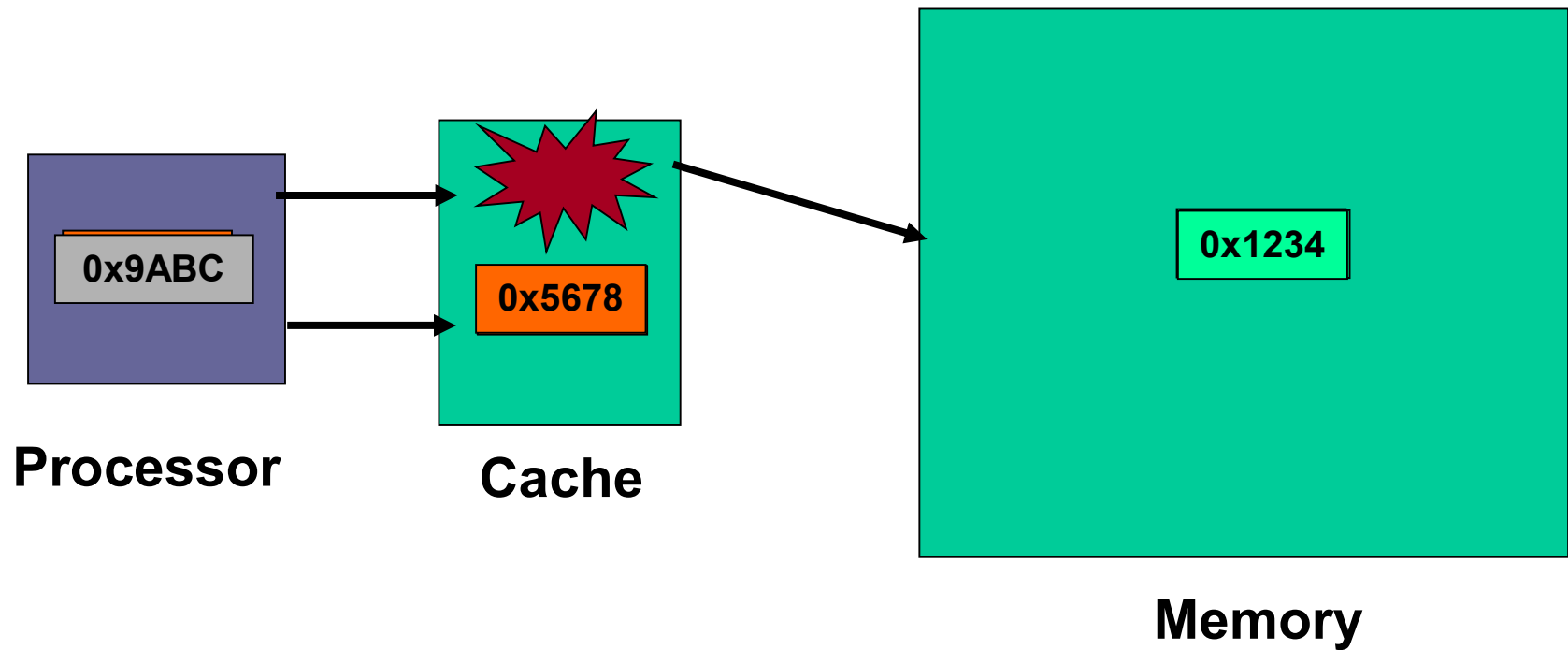


Write Buffer



- Processor: writes data into the cache and the write buffer
- Memory controller: writes contents of the buffer to memory
- Write buffer is a FIFO structure:
 - Typically 4 to 8 entries
 - Desirable: Occurrence of Writes $\ll$ DRAM write cycles
- Memory system designer's nightmare:
 - Write buffer saturation (i.e., Writes $\rightarrow$ DRAM write cycles)

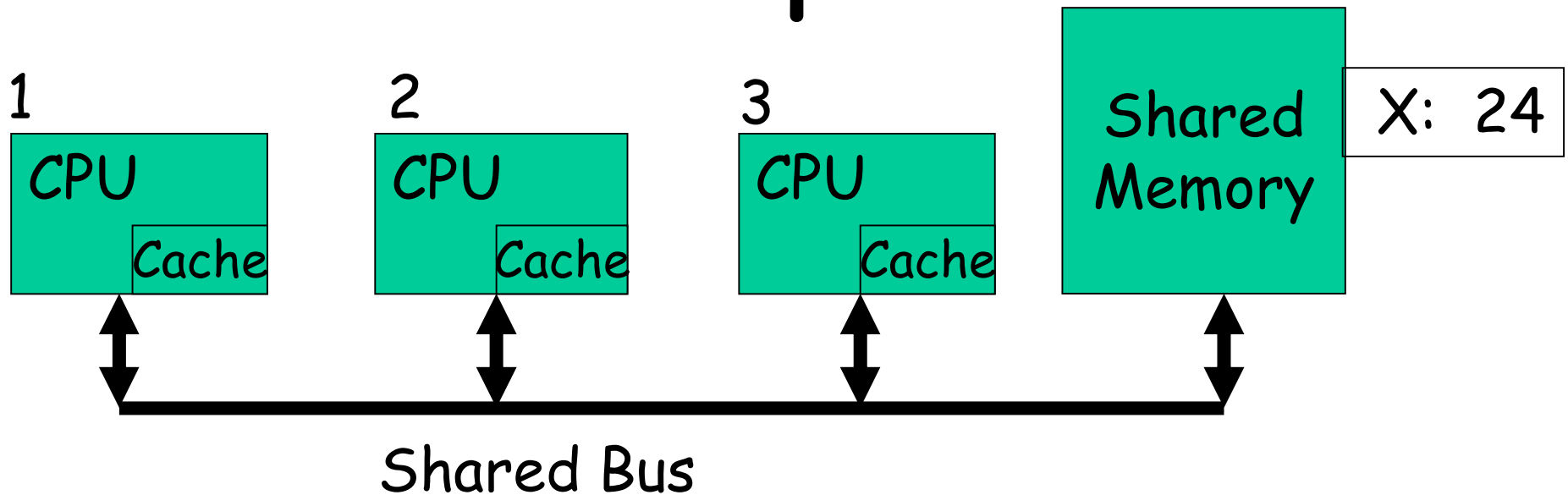
Writeback Policy



Cache Coherence Problem

- A bit called as "valid bit" is provided for each block.
- If the block contains valid data, then the bit is set to 1, else it is 0.
- Valid bits are set to 0, when the power is just turned on.
- When a block is loaded into the cache for the first time, the valid bit is set to 1.
- Data transfers between main memory and disk occur directly bypassing the cache.
- When the data on a disk changes, the main memory block is also updated.
- However, if the data is also resident in the cache, then the valid bit is set to 0.
- What happens if the data in the disk and main memory changes and the write-back protocol is being used?
- In this case, the data in the cache may also have changed and is indicated by the dirty bit.
- The copies of the data in the cache, and the main memory are different. This is called the cache coherence problem.
- One option is to force a write-back before the main memory is updated from the disk.

Example



Processor 1 reads X: obtains 24 from memory and caches
Processor 2 reads X: obtains 24 from memory and caches
Processor 1 writes 32 to X: its locally cached copy is updated
Processor 3 reads X: what value should it get?

Memory and processor 2 think it is 24
Processor 1 thinks it is 32

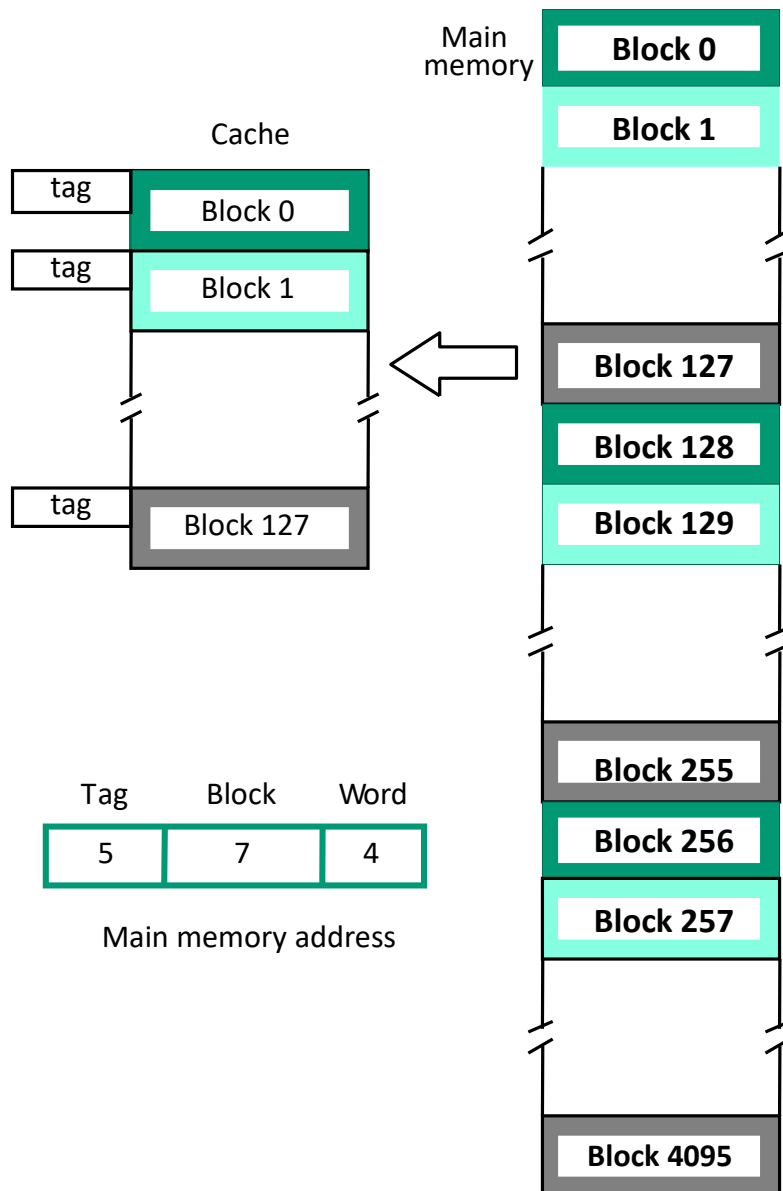
69

Notice that having write-through caches is not good enough

Mapping functions

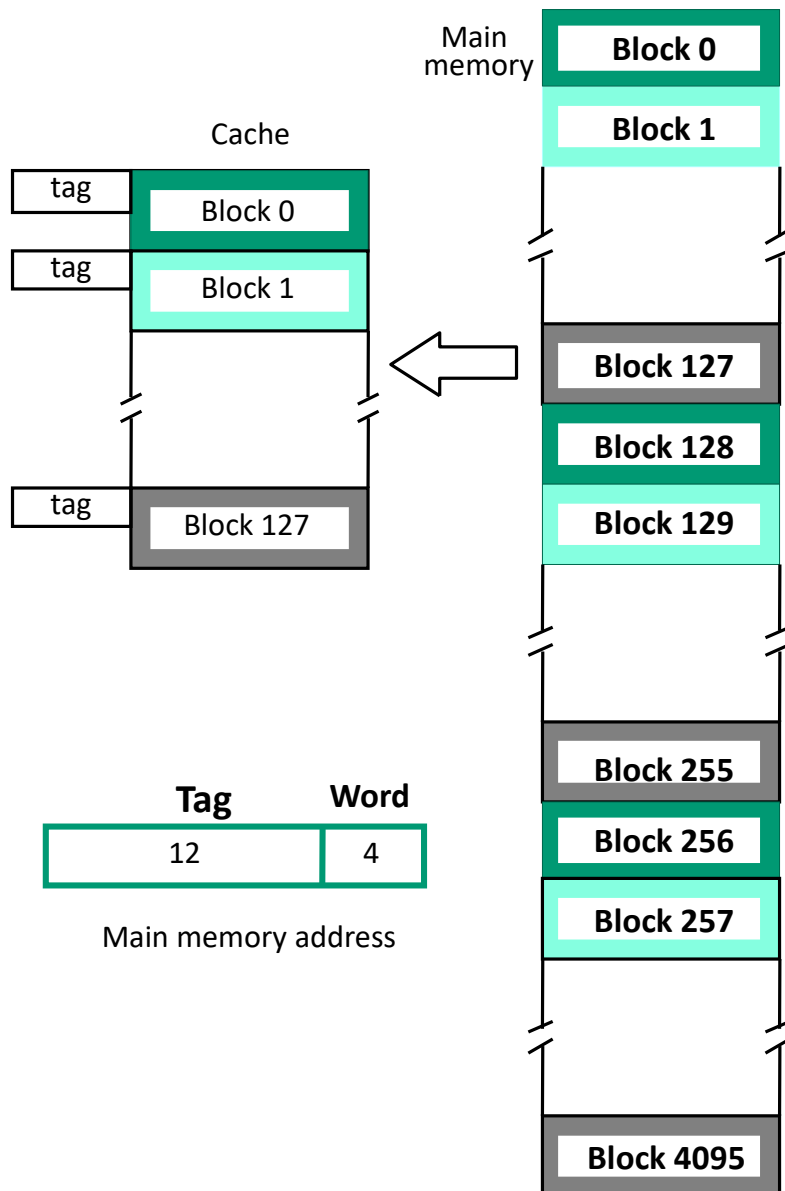
- Mapping functions determine how memory blocks are placed in the cache.
- A simple processor example:
 - Cache consisting of 128 blocks of 16 words each.
 - Total size of cache is 2048 (2K) words.
 - Main memory is addressable by a 16-bit address.
 - Main memory has 64K words.
 - Main memory has 4K blocks of 16 words each.
- Three mapping functions:
 - Direct mapping
 - Associative mapping
 - Set-associative mapping.

Direct mapping



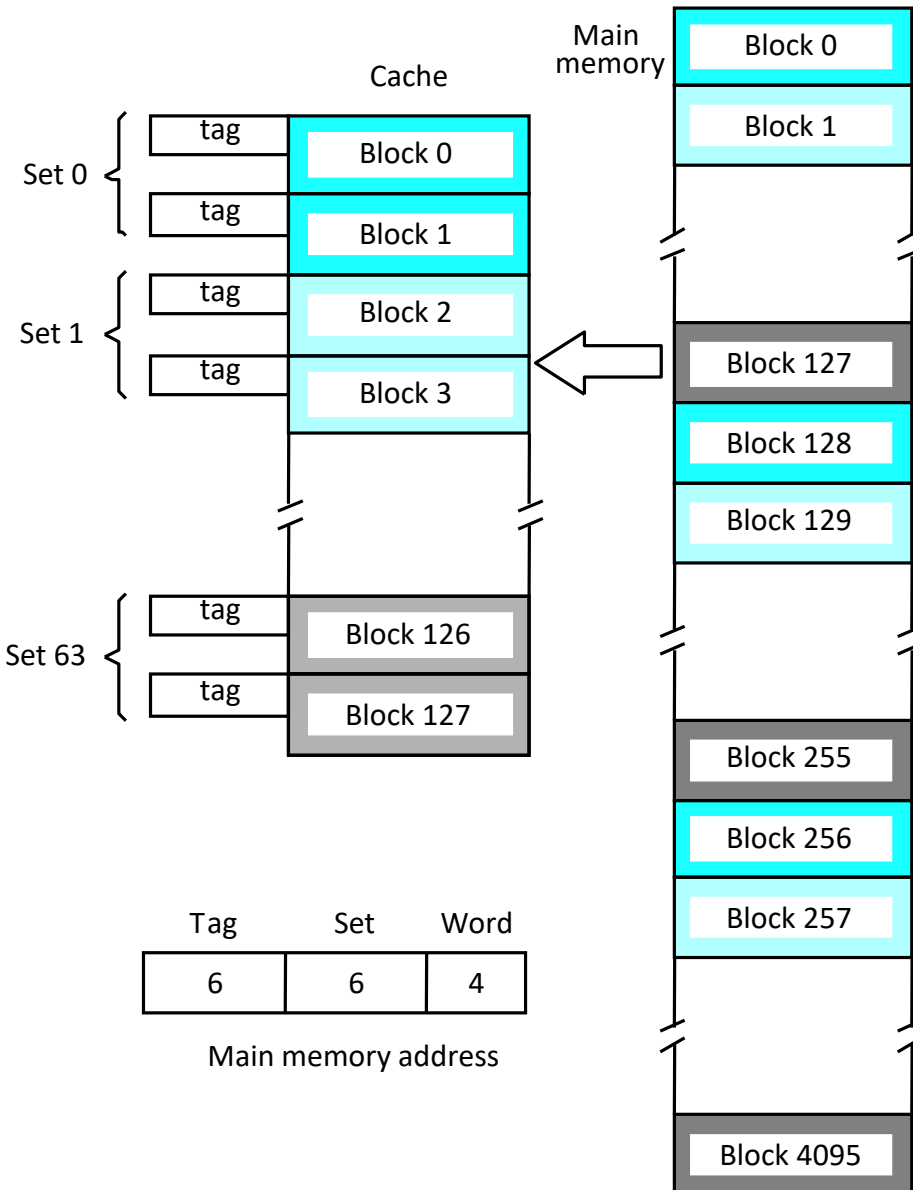
- Block j of the main memory maps to j modulo 128 of the cache. 0 maps to 0, 129 maps to 1.
- More than one memory block is mapped onto the same position in the cache.
- May lead to contention for cache blocks even if the cache is not full.
- Resolve the contention by allowing new block to replace the old block, leading to a trivial replacement algorithm.
- Memory address is divided into three fields:
 - Low order 4 bits determine one of the 16 words in a block.
 - When a new block is brought into the cache, the the next 7 bits determine which cache block this new block is placed in.
 - High order 5 bits determine which of the possible 32 blocks is currently present in the cache. These are tag bits.
- Simple to implement but not very flexible.

Associative mapping



- Main memory block can be placed into any cache position.
- Memory address is divided into two fields:
 - Low order 4 bits identify the word within a block.
 - High order 12 bits or tag bits identify a memory block when it is resident in the cache.
- Flexible, and uses cache space efficiently.
- Replacement algorithms can be used to replace an existing block in the cache when the cache is full.
- Cost is higher than direct-mapped cache because of the need to search all 128 patterns to determine whether a given block is in the cache.

Set-Associative mapping



Blocks of cache are grouped into sets.

Mapping function allows a block of the main memory to reside in any block of a specific set. Divide the cache into 64 sets, with two blocks per set.

Memory block 0, 64, 128 etc. map to block 0, and they can occupy either of the two positions.

Memory address is divided into three fields:

- 6 bit field determines the set number.
- High order 6 bit fields are compared to the tag

fields of the two blocks in a set.

Set-associative mapping combination of direct and associative mapping.

Number of blocks per set is a design parameter.

- One extreme is to have all the blocks in one set,

requiring no set bits (fully associative mapping).

- Other extreme is to have one block per set, is

the same as direct mapping.

Other Performance Enhancements

Write buffer

- Write-through:
 - Each write operation involves writing to the main memory.
 - If the processor has to wait for the write operation to be complete, it slows down the processor.
 - Processor does not depend on the results of the write operation.
 - Write buffer can be included for temporary storage of write requests.
 - Processor places each write request into the buffer and continues execution.
 - If a subsequent Read request references data which is still in the write buffer, then this data is referenced in the write buffer.
- Write-back:
 - Block is written back to the main memory when it is replaced.
 - If the processor waits for this write to complete, before reading the new block, it is slowed down.
 - Fast write buffer can hold the block to be written, and the new block can be read first.

Performance considerations

The Memory System

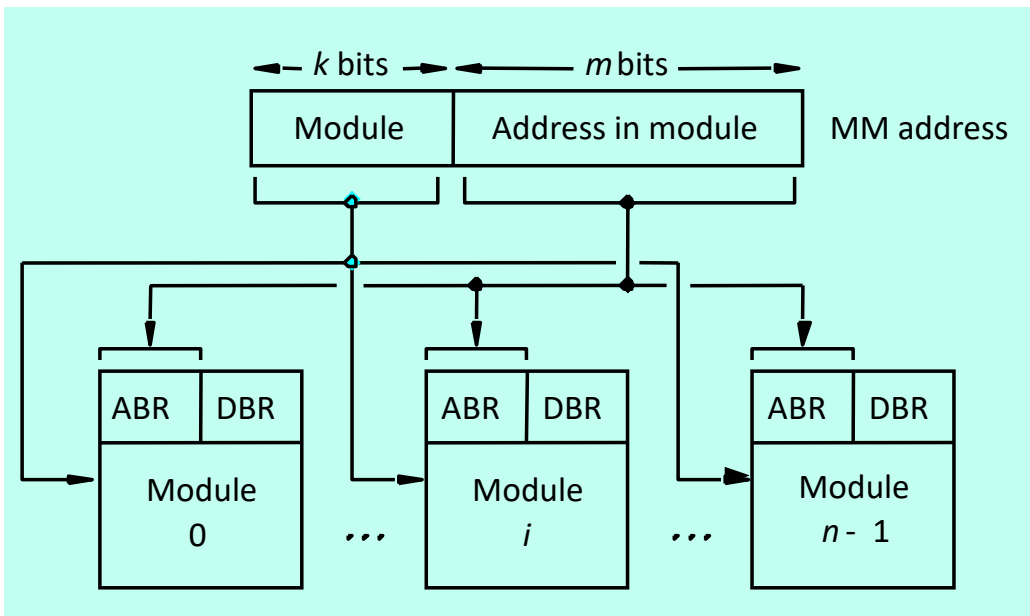
Performance considerations

- A key design objective of a computer system is to achieve the best possible performance at the lowest possible cost.
 - Price/performance ratio is a common measure of success.
- Performance of a processor depends on:
 - How fast machine instructions can be brought into the processor for execution.
 - How fast the instructions can be executed.

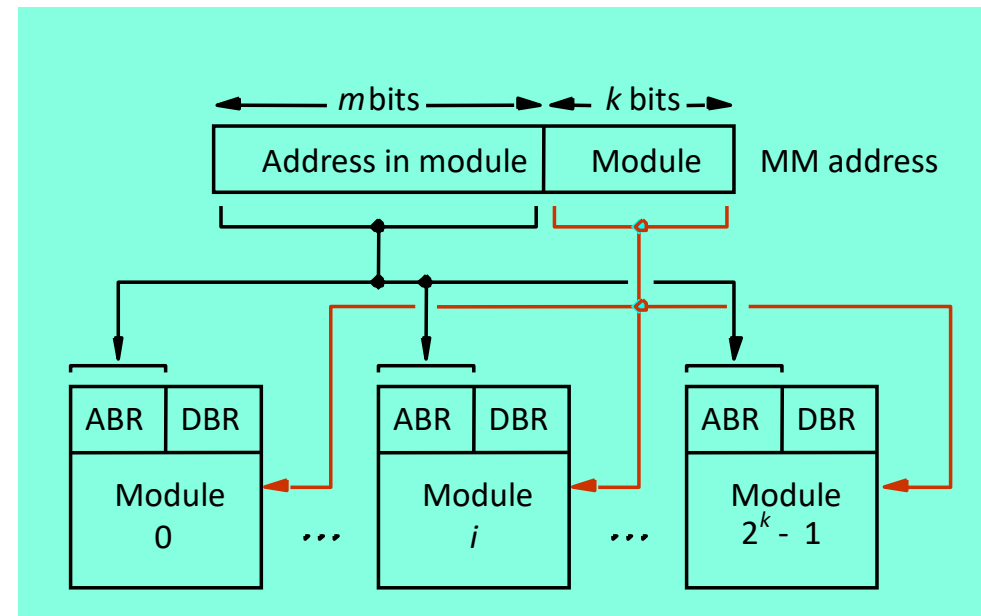
Interleaving

- Divides the memory system into a number of memory modules. Each module has its own address buffer register (ABR) and data buffer register (DBR).
- Arranges addressing so that successive words in the address space are placed in different modules.
- When requests for memory access involve consecutive addresses, the access will be to different modules.
- Since parallel access to these modules is possible, the average rate of fetching words from the Main Memory can be increased.

Methods of address layouts



- Consecutive words are placed in a module.
- High-order k bits of a memory address determine the module.
- Low-order m bits of a memory address determine the word within a module.
- When a block of words is transferred from main memory to cache, only one module is busy at a time.



- Consecutive words are located in consecutive modules.
- Consecutive addresses can be located in consecutive modules.
- While transferring a block of data, several memory modules can be kept busy at the same time.

. High Order Interleaving - In high-order interleaving, the most significant bits of the address select the memory chip.

Low Order Interleaving - In low-order interleaving, the least significant bits select the memory bank (module).

Memory Interleaving

Advantages :

- Reduced memory access latency
 - Memory Access is more efficient
- Improved memory bandwidth
 - Multiple memory modules work in parallel
 - Faster data transfer rates.
- Increased memory capacity
 - Enables use of more memory modules.

*Memory access latency is the time delay between a processor initiating a data request (read or write) and the completion of that operation

**the maximum rate at which data can be transferred to and from a computer's memory (like RAM or GPU memory) within a given time period, usually measured in bytes per second

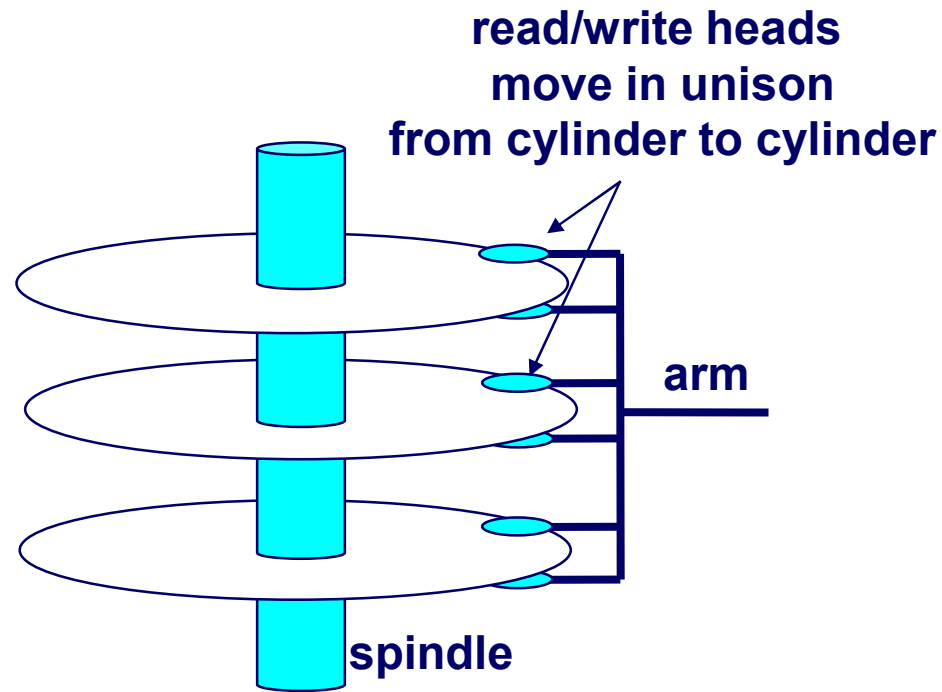
Secondary Storage

The Memory System

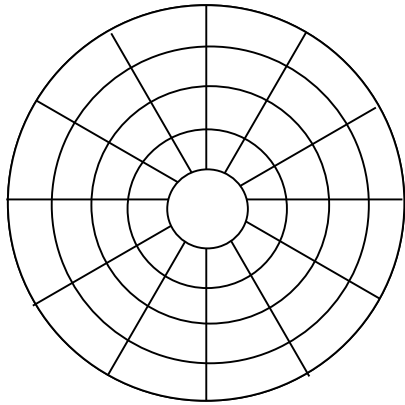
Storage Devices

- Types of storage devices in Desktop and Server systems:
 - Magnetic Disks
 - Magnetic Tapes
 - Automated Tape Libraries
 - CDs
 - DVDs
- Let's look a little bit at how these are built and what they do.

Disk Operation (Multi-Platter View)



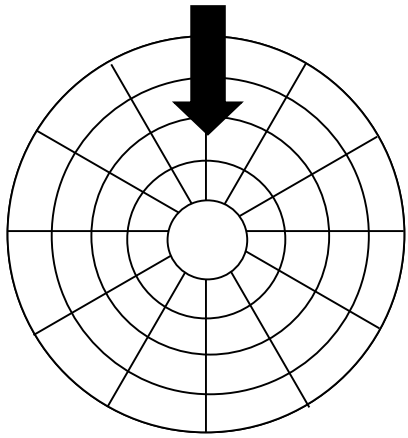
Disk Structure - top view of single platter



Surface organized into tracks

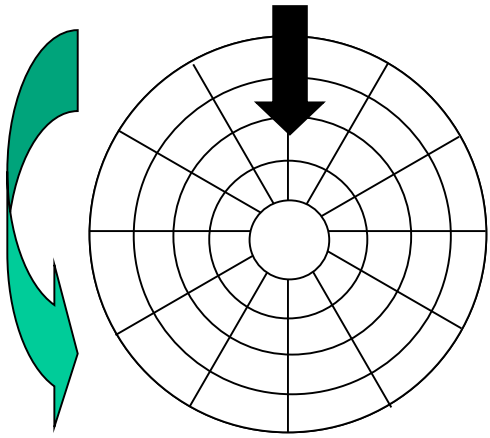
Tracks divided into sectors

Disk Access



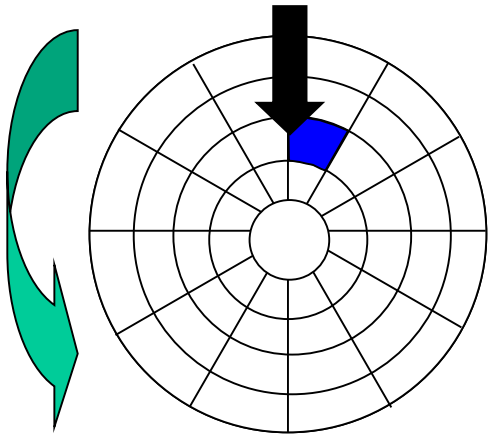
Head in position above a track

Disk Access



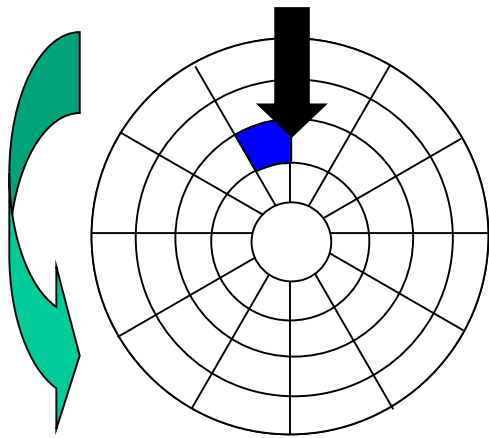
Rotation is counter-clockwise

Disk Access - Read



About to read blue sector

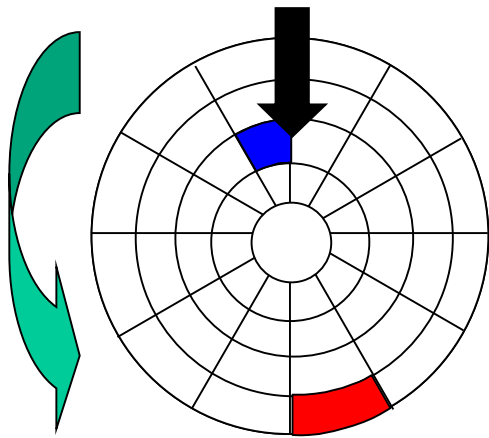
Disk Access - Read



After BLUE
read

After reading blue sector

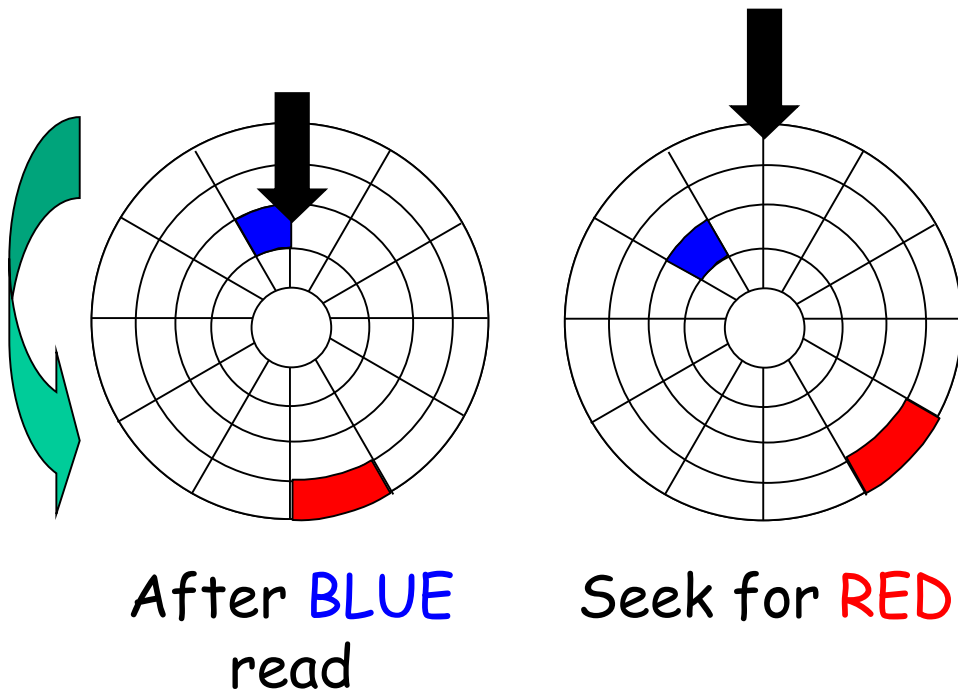
Disk Access - Read



After BLUE
read

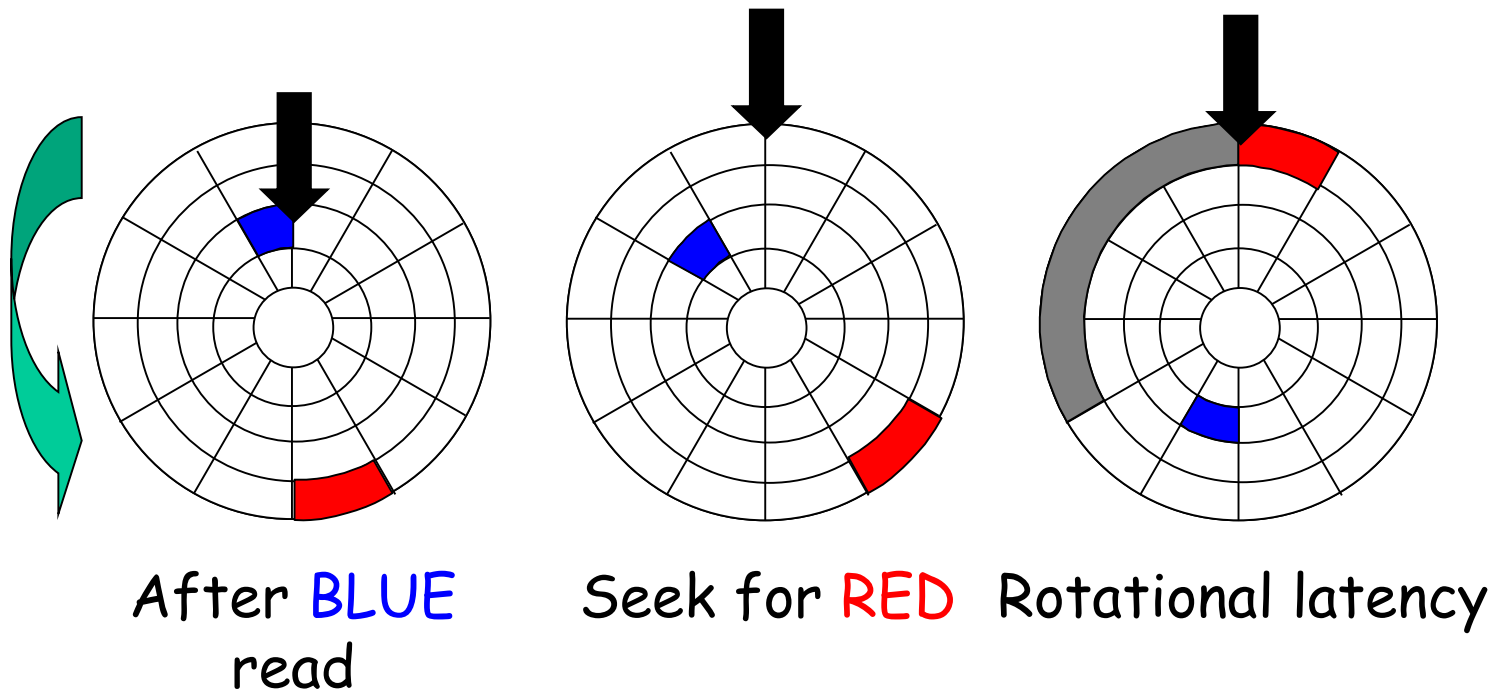
Red request scheduled next

Disk Access - Seek



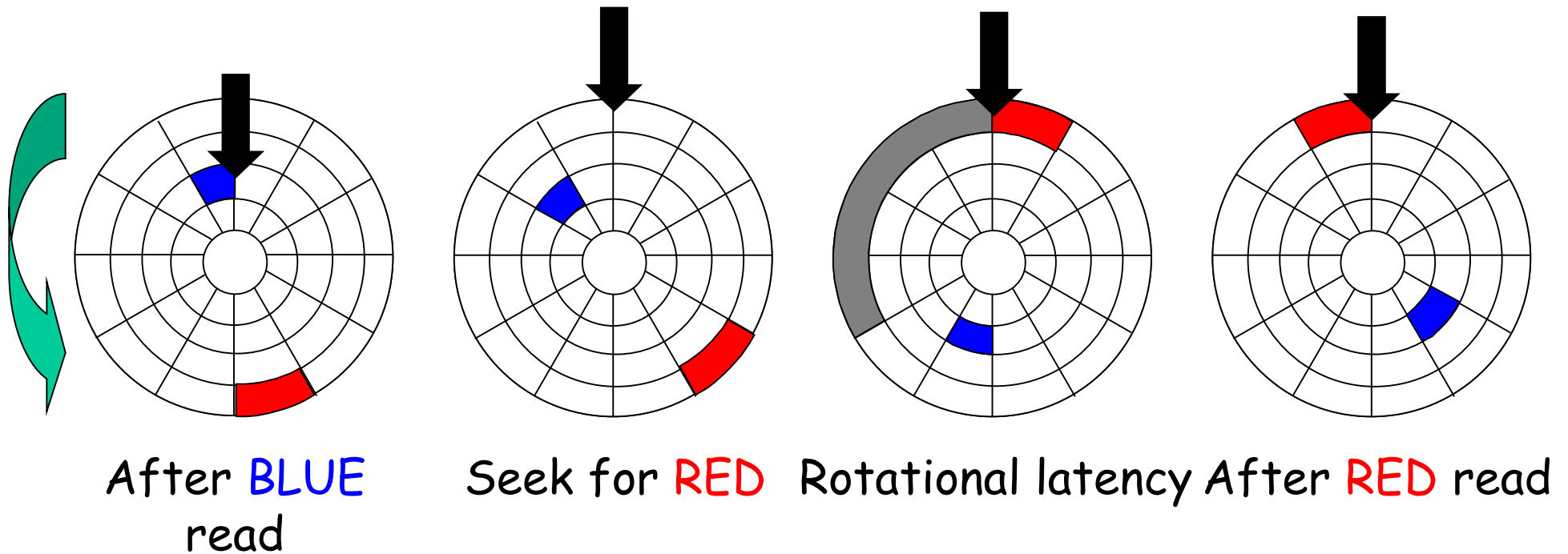
Seek to red's track

Disk Access - Rotational Latency



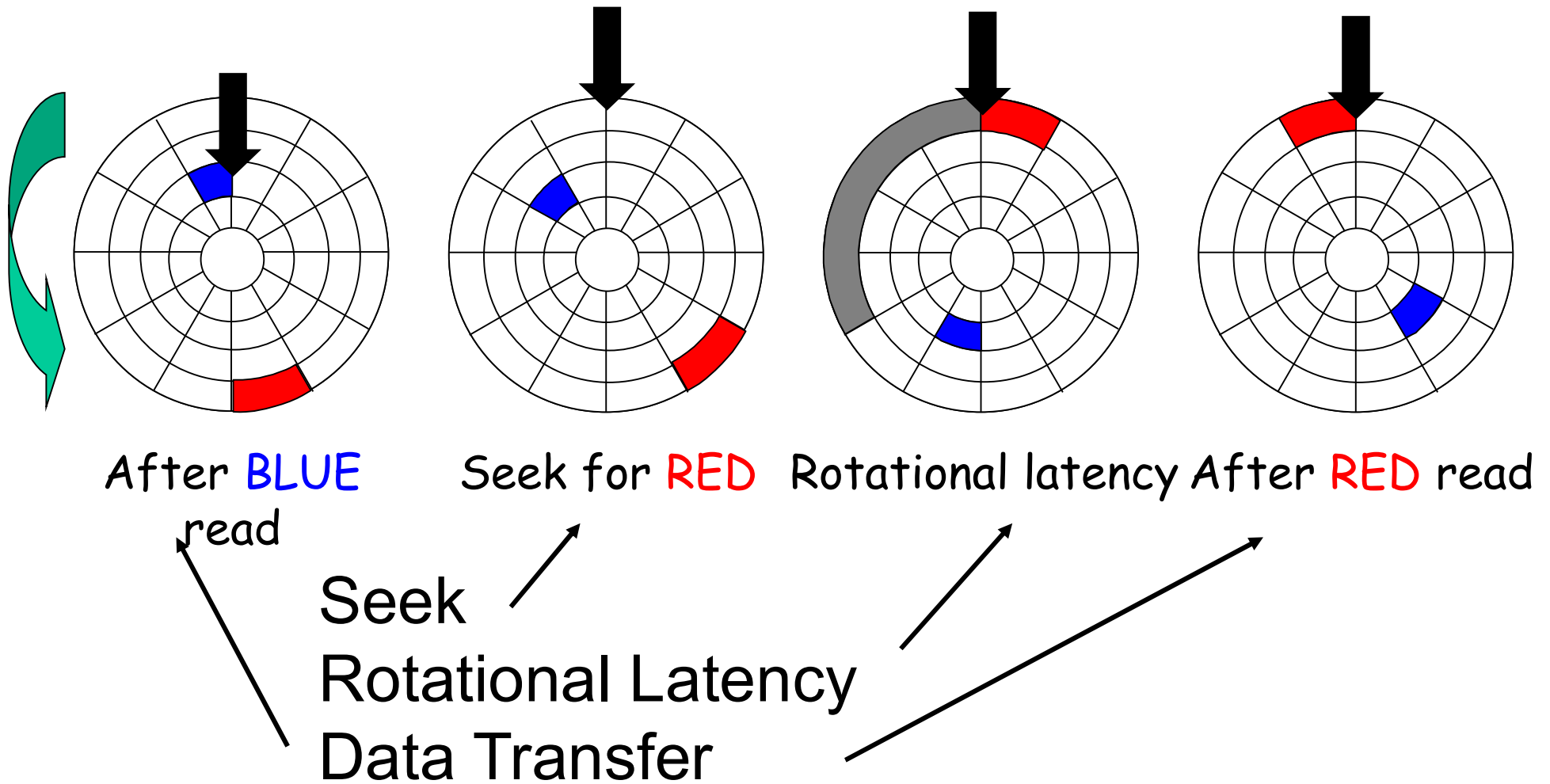
Wait for red sector to rotate around

Disk Access - Read



Complete read of red

Disk Access - Service Time Components



Disk Controller

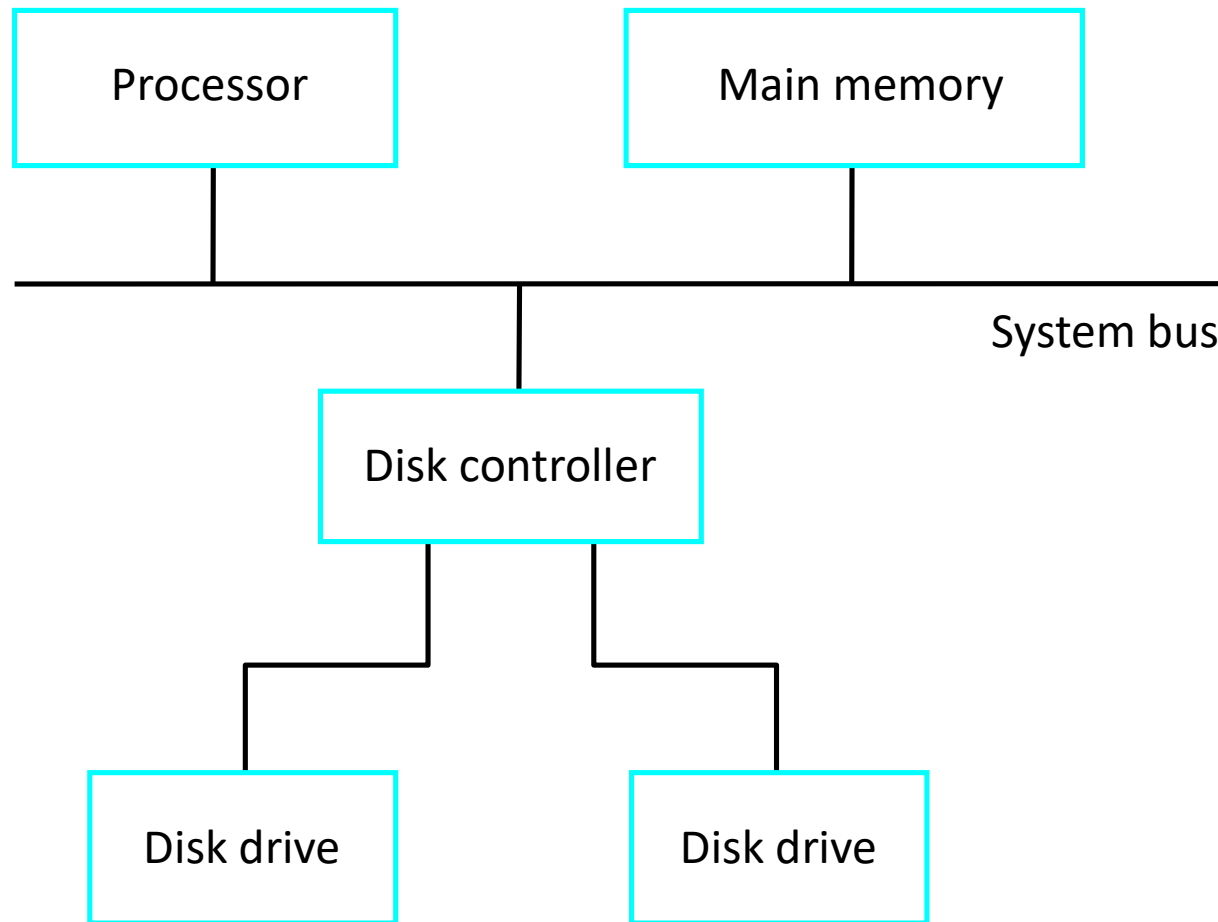


Figure 5.31. Disks connected to the system bus.⁹⁴

Disk Controller

- Seek
- Read
- Write
- Error checking

Magnetic Disk(Contd..)

- The time taken to *move the read/write head to the desired track* is called the **seek time**.
- Once the read/write head is at the right track, then the head *waits for right sector* to come under it .
 - The time taken for *desired sector of the track to come under read/write head* is called the **latency time**.
- Once the read/write head is positioned at the right track and sector, the data has to be written to disk or read from disk.
 - The rate at which *data is written to disk or read from disk* is called **data transfer rate**.
- *Access time of disk = Seek time + Latency time + Data transfer time*

Optical Disks

- CD-ROM
- CD-Recordable (CD-R)
- CD-ReWritable (CD-RW)
- DVD
- DVD-RAM

Magnetic Tape Systems

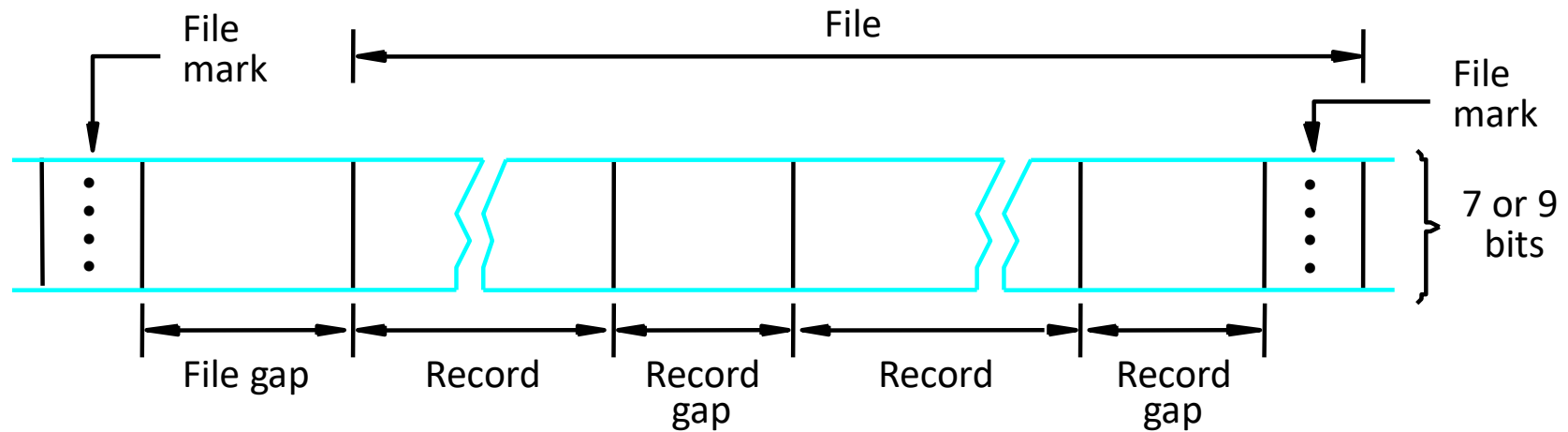


Figure 5.33. Organization of data on magnetic tape.

Magnetic Tape

- Serial access
- Slow
- Very cheap
- Backup and archive

Primary Memory(RAM)	Secondary Memory
Primary memory is computer memory that a processor or computer accesses first or directly.	Secondary memory is non-volatile and persistent in nature and is not directly accessed by a computer/processor.
fast	Slow
Expensive as compare with SM	Less expensive than primary
Capacity is less as compare with SM	Capacity large

Basic RAM size = 128 x 8

Required RAM size = 512 x 8

Text/Reference Books

TextBooks:

1. C. Hamacher, Z. Vranesic and S. Zaky, "Computer Organization", 5th edition, McGrawHill, 2017
2. W. Stallings, "Computer Organization and Architecture -Designing for Performance", 11th edition, Prentice Hall of India,2022

References:

1. D. A. Patterson and J. L. Hennessy, "Computer Organization and Design The Hardware/Software Interface", 1998
2. J .P. Hayes, "Computer Architecture and Organization", 1998
3. https://onlinecourses.nptel.ac.in/noc21_cs61/ preview

